

Quick-Start Tutorial for VMware Horizon 8

VMware Horizon 8 (2006 and later)

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Technical Introduction and Features

This *Quick-Start Tutorial for VMware Horizon 8* provides a technical overview of the VDI (virtual desktop infrastructure) and published-applications components of [VMware Horizon®](#). Published applications are offered through Microsoft Remote Desktop Session Host (RDSH). Both VDI and RDSH publishing are done through a single Horizon platform, which simplifies desktop administration and operations, and enhances user experience.

In comparison to physical desktops, delivering Horizon virtual desktops as a service enhances the security of applications and data and improves IT responsiveness, while at the same time reducing costs. The user enjoys a consistent and responsive experience across devices and locations, while maintaining IT-approved levels of customization.

Important: The exercises in this tutorial walk you through the process of setting up a VMware Horizon environment that uses a [VMware vSphere®](#) infrastructure. A vSphere infrastructure can reside either on-premises or on one or more cloud platforms, including:

- VMware Horizon® on VMware Cloud™ on AWS
- VMware Horizon® on Azure VMware® Solution (AVS)
- VMware Horizon® on Google Cloud VMware® Engine (GCVE)
- VMware Horizon® on VMware Cloud™ on Dell EMC
- VMware Horizon® on Oracle Cloud VMware® Solution

The exercises in this guide assume you are using an on-premises environment. For information about deploying Horizon on vSphere in other environments, which is beyond the scope of this guide, see the following documents:

- [Deploying VMware Horizon on VMware Cloud on Dell EMC](#)
- [Deploying VMware Horizon on VMware Cloud on AWS](#)
- [Deploying VMware Horizon on Azure VMware Solution](#)
- [Google Cloud VMware Engine Documentation](#)
- [Oracle Cloud VMware Solution Documentation](#)

This tutorial does not include exercises for using the Horizon Cloud Service, which is a VMware-managed virtual desktop and application solution that provides desktops as a service using a Microsoft Azure or IBM Cloud public cloud infrastructure:

- VMware Horizon® Cloud Service™ on Microsoft Azure
- VMware Horizon® Cloud Service™ on IBM Cloud

A separate quick-start tutorial is available for Horizon Cloud Service: [Quick-Start Tutorial for VMware Horizon Cloud Service on Microsoft Azure](#).

This tutorial is provided to help you evaluate VMware Horizon. The first chapter provides an overview of the key VDI and RDSH features. Subsequent chapters contain exercises to guide you through the basic installation and initial configuration processes, and to explore key features and benefits.

Note: This tutorial is designed for evaluation purposes only. It uses the minimum required resources for a basic deployment and does not explore every feature. Do not use this evaluation environment as a template for a production environment. For information beyond the considerations of this tutorial, see [VMware Horizon Documentation](#).

This tutorial is intended for IT administrators, architects, engineers, and product evaluators who want to install Horizon and deploy a VDI environment. Both current and new users can benefit from using this tutorial. You should be familiar with VMware vSphere and [VMware vCenter Server](#)[®]. Familiarity with other technologies is also helpful, including networking and storage in a virtual environment, Active Directory, identity management, directory services, and RSA SecurID.

Advantages of Using VMware Horizon

VMware Horizon is a centralized desktop virtualization solution that enables organizations to deliver virtualized desktop services and applications to employees as a managed service. Horizon has advantages for both end users and IT administrators:

- End users are no longer restricted to one specific machine, and can access their system and files across many supported devices and locations.
- As an IT administrator, you can use Horizon to simplify and automate the management of desktops and applications, and you can securely deliver desktops as a service to users from the data center or cloud. You can quickly create virtual desktops on demand based on location and profile.

A single administration console provides detailed levels of control, allowing you to customize the end-user experience, access, and personalization to support corporate policy. End users get a familiar, personalized environment that they can access from any number of devices anywhere throughout the enterprise or from remote locations. And as an administrator, you have centralized control, efficiency, and security by storing desktop data in the data center.

Packaging and Licensing

Horizon is available as a perpetual entitlement or as a subscription service. For more information, see the [Compare section of the Horizon product page](#).

Also see the [Horizon Universal License page](#). The Horizon universal license provides a single subscription license for all Horizon products.

Features

Horizon offers many features that help IT efficiently deploy and scale full-featured, personalized digital workspaces from a single control plane with rapid provisioning, automation, and simplified management. For an overview of the key capabilities, see [What Is VMware Horizon?](#)

Components and Architecture

Horizon contains key components and integrated products that work together. The following diagram shows the logical architecture of a typical Horizon implementation.

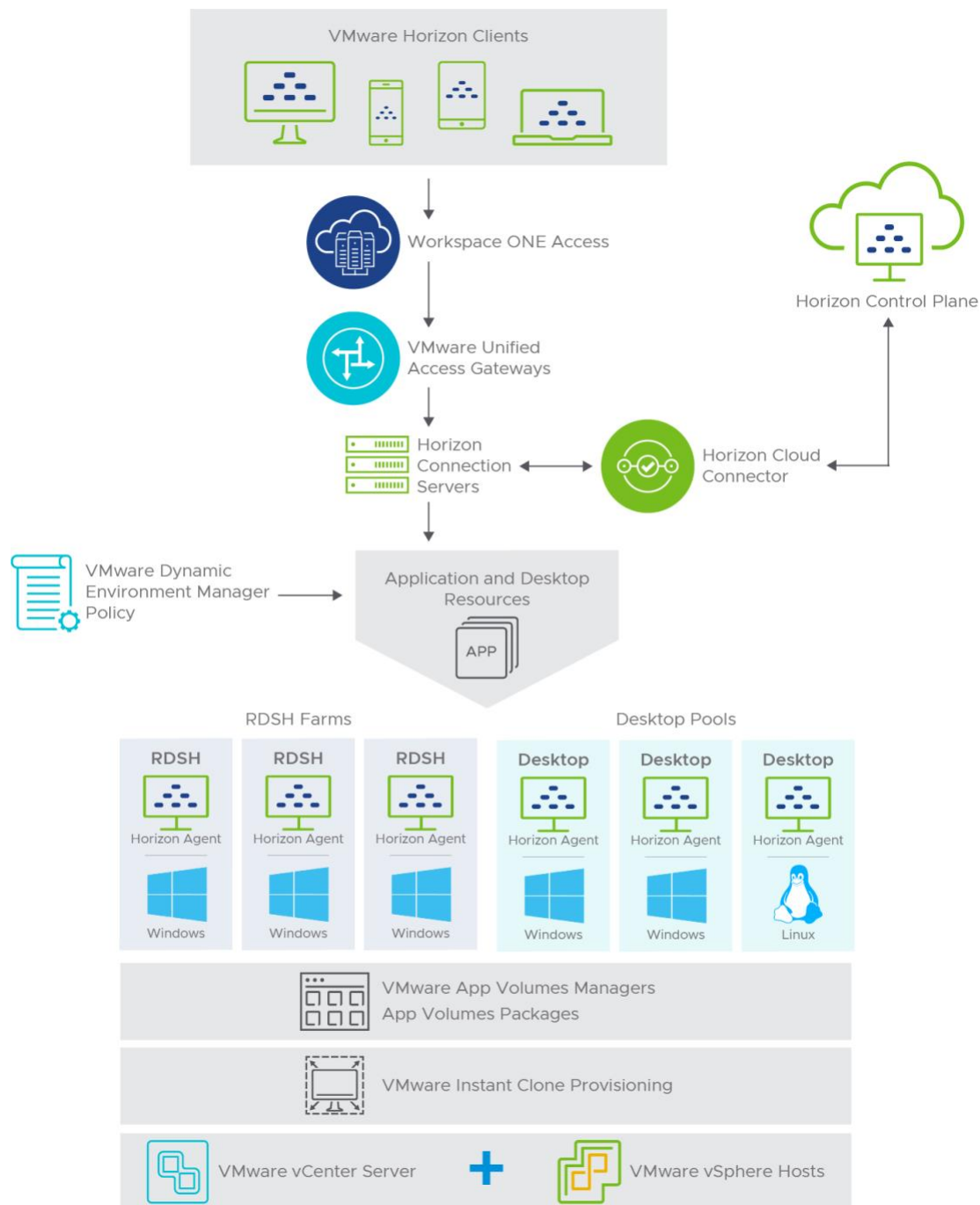


Figure 1: Logical Architecture for VMware Horizon

This figure shows how Horizon components work together to provide access to virtual desktop pools, RDSH desktop and application pools, and more. From the top of the diagram down, the components are:

Horizon Clients – VMware Horizon® Client for Windows, macOS, iOS, Linux, Chrome, or Android is installed on every endpoint so that end users can access their virtual desktops and published applications from a variety of devices such as smartphones, zero clients, thin clients, PCs, laptops, and tablets.

Horizon Client software can be downloaded for free from app stores or from VMware, or users can open a browser and enter the server URL to use the VMware Horizon® HTML Access™ web client, which does not require installing any software on the client device.

Workspace ONE Access (optional) – [VMware Workspace ONE Access™](#) (formerly called VMware Identity Manager) is a solution that provides application provisioning, a self-service catalog of applications and virtual desktops, conditional access controls, and single sign-on (SSO) for software as a service (SaaS), web, and cloud resources. When Workspace ONE Access is used to integrate Workspace ONE with Horizon, users can use the Workspace ONE Intelligent Hub app to launch their Horizon desktops and published apps.

Unified Access Gateway – [VMware Unified Access Gateway™](#) (formerly called VMware Access Point) provides a secure gateway that allows users to access their desktops and applications from outside a corporate firewall. Unified Access Gateway appliances typically reside in a demilitarized zone (DMZ) and act as a proxy host for connections inside your trusted corporate network.

Although Unified Access Gateway is included with VMware Horizon, this quick-start guide does not include exercises for deploying and configuring the Unified Access Gateway appliance. For that information, see [Deploying VMware Unified Access Gateway: VMware Workspace ONE Operational Tutorial](#).

Horizon Connection Servers – Users connect to a Connection Server, often through a Unified Access Gateway, to access their virtual desktops and native, virtual, or RDSH-based applications. The Connection Server provides the following management capabilities:

- Authenticating users
- Entitling users to specific desktops and applications
- Managing local and remote desktop and application sessions
- Establishing secure connections between users and desktops or applications
- Enabling single sign-on
- Setting and applying policies
- Managing an instant-clone engine

Horizon Control Plane and Cloud Connector (optional) – One key concept in a Horizon environment design is the use of pods. A pod is made up of a group of interconnected Connection Servers that broker connections to desktops or published applications. Horizon Control Plane is a cloud-based service that unifies and simplifies management across pods, providing monitoring as well as image, application, and lifecycle management. In addition, a global entitlement layer connects Horizon pods, letting end users access their desktop in any connected pod or cloud.

Horizon Cloud Connector is a virtual appliance that you pair with a Connection Server in an on-premises pod so that the pod can be connected to the Horizon Control Plane. This pairing also requires the use of subscription licensing.

Dynamic Environment Manager (optional) – [VMware Dynamic Environment Manager™](#) (formerly called User Environment Manager) lets administrators configure user-specific Windows desktop and application settings. The settings are applied in the context of the client device, location, or other conditions. Policies are enforced when users log in or when they launch an app, reconnect, or when some other triggering event occurs. Settings are also available for configuring folder redirection to store personal user data, including documents, pictures, and so on.

For a quick-start tutorial on Dynamic Environment Manager, see [Quick-Start Tutorial for User Environment Manager](#).

Desktop and application resources – These Windows resources are provided by single-session desktop pools or shared-session RDSH server farms. If Dynamic Environment Manager is used, profile and policy settings are also applied. If App Volumes is used, App Volumes application packages are attached and used instead of or in addition to natively installed applications.

RDSH server farms and virtual desktop pools – The virtual machines in these pools are created from a golden VM image. The Horizon Agent, which you install in the virtual desktop or RDSH server, communicates with Horizon Client on the end user's device to determine which applications and desktops to provide to the user. Horizon Agent also provides features such as connection monitoring, client drive redirection, VMware integrated printing, and access to locally connected USB devices.

App Volumes managers and packages (optional) – [VMware App Volumes™](#) uses application containers called App Volumes packages, which are virtual disks that contain all of the components that are required to run an application, such as executables and registry keys. When a package is deployed, it is available for use within seconds without end-user installation. Applications can be deployed once to a single central file and accessed by thousands of desktops. This simplifies application maintenance, deployment, and upgrades.

For a quick-start tutorial on App Volumes, see [Quick-Start Tutorial for VMware App Volumes 4 Simplified Application Management](#).

Instant-clone provisioning – This is a vSphere-enabled technology for cloning desktops and RDSH servers. It allows administrators to quickly create VMs that share virtual disks with a golden image, conserving disk space and simplifying the management of OS patches and updates—no separate server or database required. For instant-clone virtual desktops, the operating system can be either a Windows or a Linux operating system.

With Horizon 8, instant-clone smart provisioning capabilities have been added, which improve Horizon instant-clone functionality to help reduce storage requirements and costs. For more information, see the [Horizon 8 Smart Provisioning video](#).

vCenter Server and vSphere hosts – The vSphere product family includes VMware ESXi™ hosts and VMware vCenter Server, and it is designed for building and managing virtual infrastructures. The vCenter Server system provides key administrative and operational functions, such as provisioning, cloning, and VM management features, which are essential for VDI. For more information, see [VMware vSphere Documentation](#).

Sketch of the Quick-Start Steps

After completing the exercises in this guide, you will have a small Horizon environment with several published applications and a virtual desktop pool. The applications are delivered using RDSH servers. Both the server farm and the desktop pool are created using instant-clone provisioning.

Below is a graphical summary of your accomplishments by the time you reach the end of this guide.

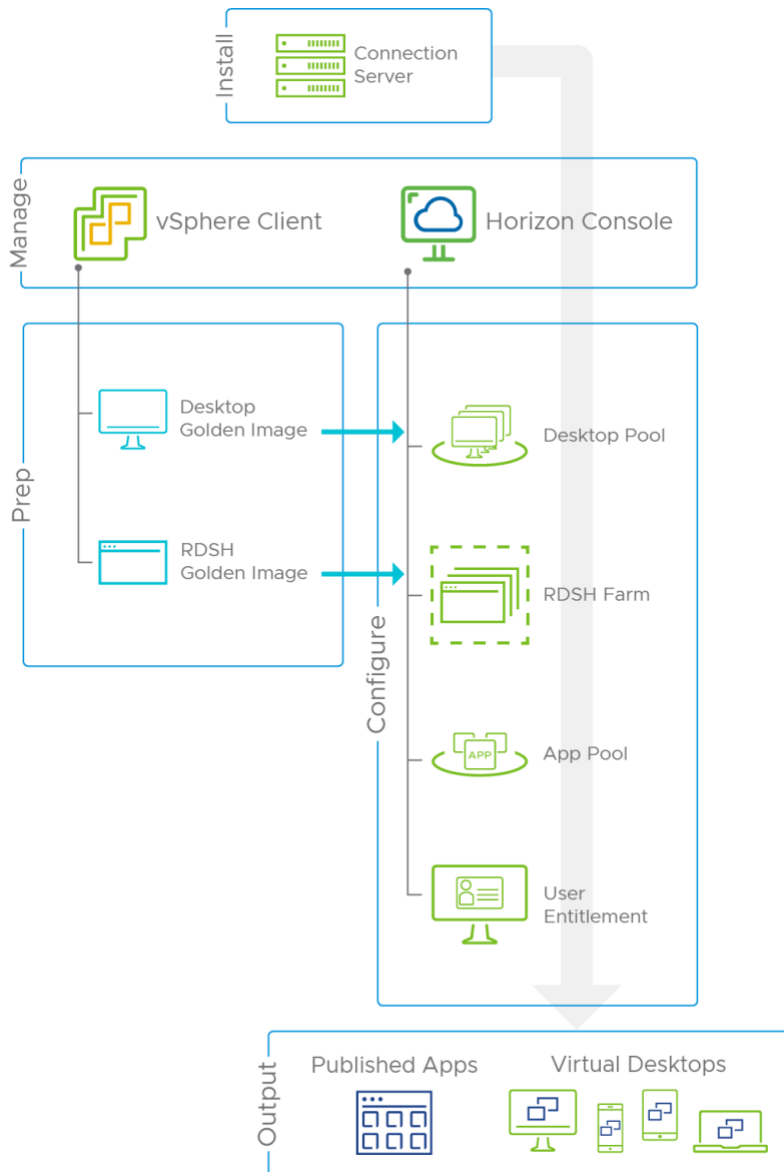


Figure 2: Workflow for Creating Virtual Desktops and Published Apps with Horizon

1. **Install** – You begin by installing the server software (Horizon Connection Server) that manages sessions between users and their virtual desktops and published apps and that enables automation of desktop and RDSH server creation. The web-based Horizon Administrative Console is included when you install a Connection Server.
2. **Prep** – Next, you use vSphere Web Client to create golden images of the Windows desktop and the RDSH server and that will host applications. Besides configuring the operating system on these VMs, you will install the agent software (Horizon Agent) that communicates with the Horizon servers and the clients.
3. **Configure** – Once these Horizon infrastructure elements are set up, delivering apps and desktops is very simple. For virtual desktops, you will use the Horizon Console to create a desktop pool from the golden image, which automatically clones the number of desktops you specify.
4. **Configure** – For published applications:
 - a. Use the Horizon Console to create an RDSH server farm from the golden image, which automatically clones the number of servers you specify.
 - b. Publish one or multiple application pools with one trip through the Add Application Pool wizard.
5. **Configure** – After the desktop and application pools are ready, use the Horizon Console to entitle users or groups to the apps and desktops.
6. On client devices, you will download the free Horizon Client software from app stores or from VMware to install on iOS, Android, Chromebook, Windows, macOS, or Linux clients, or just open a browser and enter the server URL to use the HTML Access web client.

How to Install Horizon

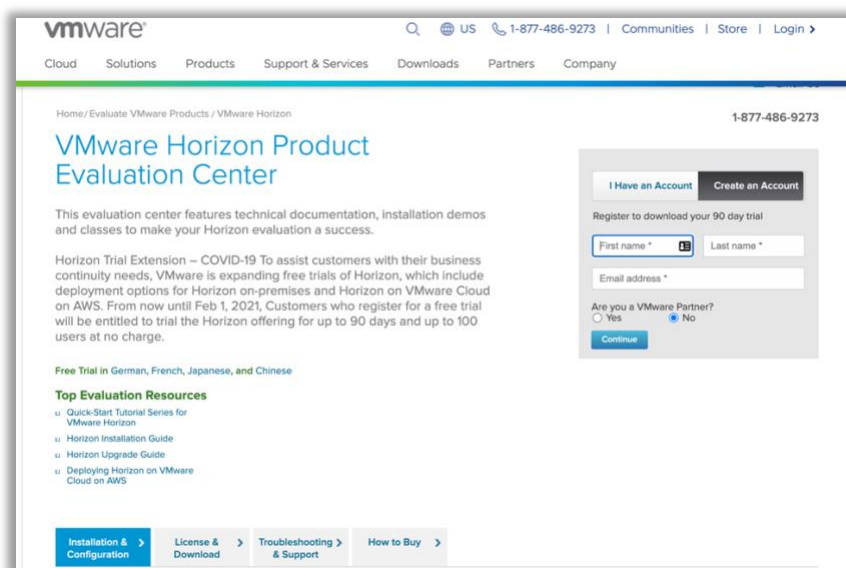
The exercises in this installation chapter are sequential and build on one another. For this reason, be sure to complete each exercise before moving on to the next.

Download Horizon Installers

If you have already purchased Horizon, you can download the installers (installation files) from the [Download VMware Horizon](#) page. This exercise shows you which installers to download and how to download the installers from the VMware Product Evaluation Center, which gives you a free 90-day trial.

To download the installers:

1. On any web browser, navigate to the [VMware Horizon Product Evaluation Center](#), and log in. If you do not already have an account, you can create one here.



2. If prompted, register for the product.
3. On the **License & Download** tab, scroll down to the **License Information** section, and make a note of the Horizon Enterprise license.
You will use this license key after you run the installer.
4. Continue scrolling down to **Download Packages** section, expand **VMware Horizon Enterprise Binaries**, if necessary, and download the following packages:
 - Horizon Connection Server (64-bit)
 - Horizon Agent (64-bit)
 - Horizon GPO Bundle

5. If you do not already have a vSphere environment set up, scroll down and expand the **Hypervisor and Management Server Binaries** section, and download the following packages:
 - VMware vCenter Server Appliance
 - VMware vSphere Hypervisor (ESXi ISO) image
6. On the client device you plan to use—desktop computer, laptop, Chromebook, tablet, or phone—navigate to the [Download VMware Horizon Clients](#) page and download the appropriate client installer for that device type. The exercise in this guide provides instructions for accessing virtual desktops and published apps using a Windows device.

Verify Infrastructure Requirements Are Met

Before you begin the installation exercises in this guide, make sure that your environment meets the following infrastructure requirements:

- **VMware vSphere and vCenter Server** – Before you can perform the exercises in this guide, you must have a [VMware vSphere](#) infrastructure that contains at least one VMware [ESXi host](#) and one [VMware vCenter Server](#) instance. This guide does not provide instructions for installing these vSphere components. For instructions, see the [vSphere Product Documentation](#).
- **Active Directory domain controller** – The authentication infrastructure for your setup must include Active Directory, DNS, and DHCP. The Connection Server joins to Active Directory and sets up a lightweight directory service instance for the storage of Horizon configuration information.
- **TLS/SSL certificate** – (Optional) By default, Horizon servers include a self-signed certificate that can be used for testing purposes. For a production environment, we recommend that you replace the self-signed certificate with an approved certificate signed by a certificate authority, a trusted entity that issues digital certificates verifying another digital entity's identity on the Internet.
- **SQL database server** – This is the database server on which you will create the Events database, which records actions that occur on the Horizon servers. For the example in this exercise, we used Microsoft SQL Server 2019.

For the most up-to-date information about supported databases, see the VMware Product Interoperability Matrixes at http://www.vmware.com/resources/compatibility/sim/interop_matrix.php. For **Solution/Database** Interoperability, after you select the product and version, for the **Add Database** step, to see a list of all supported databases, select **Any** and click **Add**.

If you need to create a VM to host a database server, you can follow the instructions in the section [Create VMs for the Connection Server and Database Server](#).

Note: You can download and install [Microsoft SQL Server Management Studio Express](#) with Advanced Services to get both database and management tools, or use an existing SQL server in your environment.

- **Network and storage** – VMware recommends a network connection speed of at least 1 Gbps between all the required Horizon components and desktops. The standard vSphere network is sufficient for a Horizon environment. Similarly, for a Horizon test environment, you can use the local storage on a vSphere ESXi host if you do not have separate storage or VMware vSAN™.

Create VMs for the Connection Server and Database Server

For the exercises in this guide, you must have VMs on which to install the Connection Server and the Microsoft SQL Server database server. For this purpose, you create a VM template and clone it to create the required VMs for the server components.

Note: If you are already using a vSphere environment in production and have production VMs with Windows Server installed, you might be able to use those or clone them, as long as they meet the virtual hardware requirements. If not, you can use the procedure in this exercise.

Outline for creating a server VM:

1. Using the step-by-step instructions in the guide [Creating an Optimized Windows Image for a VMware Horizon Virtual Desktop](#), complete the tasks in the chapter called *Initial VM Creation*; specifically:
 - a. *Create a Virtual Machine* section, using the following virtual hardware specifications:
 - Guest OS** – Microsoft Windows Server 2019
 - CPU** – 4
 - Memory** – 4 GB
 - New Hard Disk** – 40 GB
 - New SCSI Controller** – LSI Logic SAS
 - Network Adapter Type** – VMXNET3
 - b. *Install Windows* section
 - c. *Install VMware Tools* section
 - d. *Install .NET Framework 3.5* section
 - e. *Update Windows* section
2. Clone the virtual machine you just created to a virtual machine template, which you can use to easily create the server VMs you need.
For instructions, see the vSphere documentation topic [Clone a Virtual Machine to a Template](#).
3. Deploy the Connection Server VM and, if necessary, the database server VM from the template you just created.
For instructions, see the vSphere product documentation topic [Deploy a Virtual Machine from a Template](#).
Important: These VMs must have an IP address that does not change. In an IPv4 environment, configure a static IP address for each VM. In an IPv6 environment, machines automatically get IP addresses that do not change.

Now that you have VMs for your Horizon servers, you can copy the Horizon Connection Server installer to the VM and run the wizard.

Install the Horizon Connection Server

The Connection Server acts as a broker for client connections by authenticating and directing incoming user desktop requests. When you install the Connection Server, the Horizon Console is installed as well. The Horizon Console is the web-based interface for the management, provisioning, and deployment of virtual desktops. As an administrator, you can centrally manage thousands of virtual desktops from a single Horizon Console.

Prerequisites for Connection Server Installation

To perform this exercise, you will need the following:

- **User account** – When you log in to the OS to run the installer, the account you use must have administrative privileges.
- **Installer** – If necessary, you can download the installer from the [Download VMware Horizon](#) page or the [VMware Horizon Product Evaluation Center](#). You must download and copy the installer file to the Connection Server VM, or, alternatively, you can copy it to a location accessible to the system.
- **Connection Server VM** – Verify that you have performed the exercise [Create VMs for the Connection Server and Database Server](#) so that you have a server VM with the required virtual hardware, a supported Windows OS, and a static IP address.

For more information about virtual hardware requirements, see [Hardware Requirements for Horizon Connection Server](#) in the *Horizon Installation* guide. For a complete list of supported operating systems, see the VMware Knowledge Base article [Supported Operating Systems and MSFT Active Directory Domain Functional Levels for VMware Horizon 8 2006 \(78652\)](#).

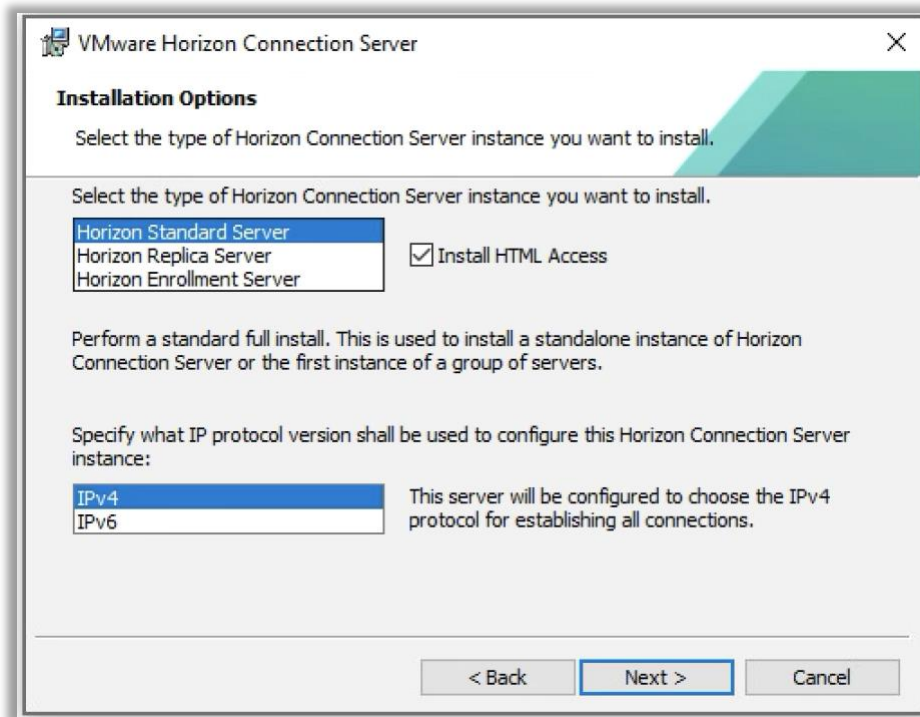
Run the Horizon Connection Server Installation Wizard

You install the Horizon Connection Server in the same manner as you would other Windows applications.

To install Horizon Connection Server:

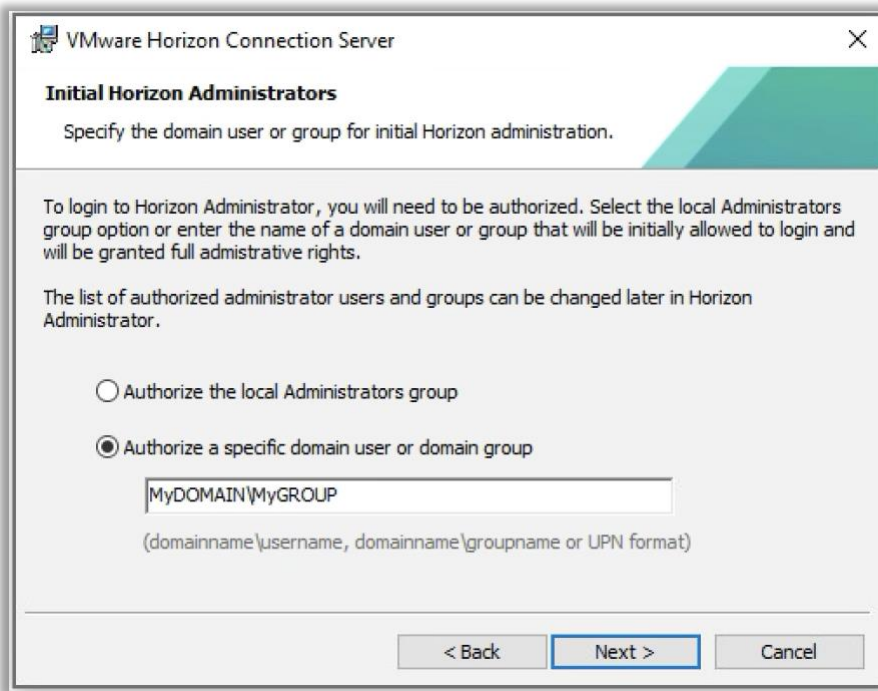
1. Navigate to the Connection Server installation file that you downloaded earlier, and double-click the file to start the installation wizard.
The file name has the format `VMware-Horizon-Connection-Server-x86_64-x.x.x-build_number.exe`, where `x.x.x` is the version number.
2. If prompted to allow the app to make changes to your device, click **Yes**.
The Welcome page of the wizard appears.
3. Follow the prompts to the Installation Options page, and select the following options before clicking **Next**:
 - **Horizon Standard Server** – This is the default. For information about the other types of Connection Servers you can install after first installing a standard server, see [Installing the Horizon Connection Server Software](#) in the *Horizon Installation* guide.
 - **Install HTML Access** – This is the default. HTML Access is the web-based Horizon Client, which lets you access virtual desktops and applications that do not have any client software installed.

- **IP protocol version** – You must install all VMware Horizon components, including the Horizon Agent on desktop VMs and RDSH server VMS, with the same IP version.



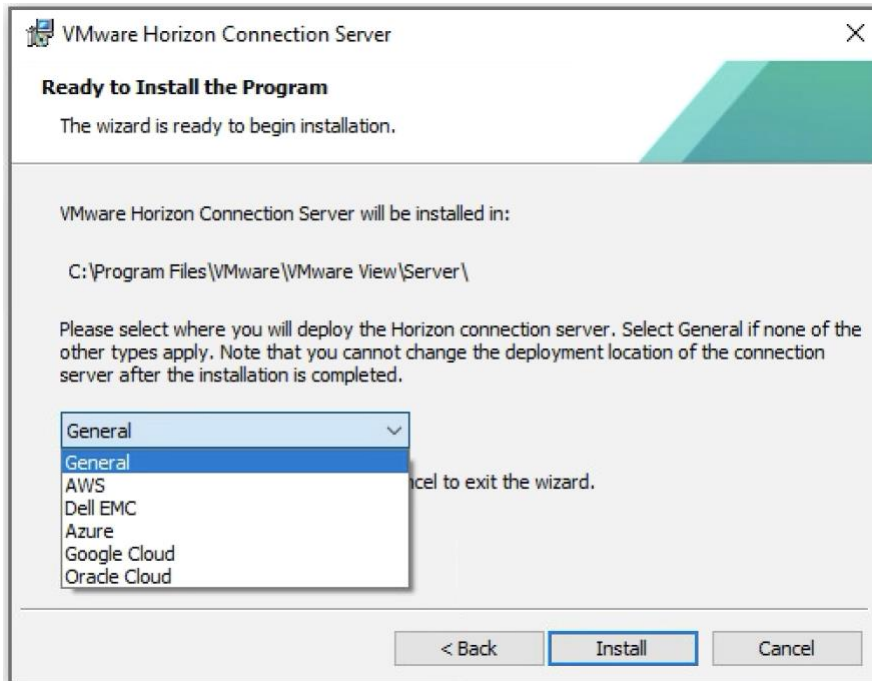
4. On the Data Recovery page, enter the password you want to use for recovering data backups of the Connection Server.
5. On the Firewall Configuration page, accept the default, which is **Configure Windows Firewall automatically**.

6. On the Initial Horizon Administrators page, for this exercise and for simplicity, we recommend authorizing an Active Directory domain group.



7. On the User Experience Improvement Program page, you can deselect the **Join the VMware Customer Experience Improvement Program** option to opt out of the program.

8. On the Ready to Install page, leave the default for the drop-down list, which is **General**, to indicate that you are deploying the Connection Server in an on-premises environment, and click **Install**.



9. On the Installer Completed page, click **Finish**.
10. Launch the Horizon Console by using one of the following methods:
 - If you are logged in to the server on which you installed the Connection Server, either double-click the Horizon Administrator Console desktop shortcut or open a browser and enter the following URL:
`https://localhost/admin/`.
 - If you are accessing the Horizon Console from a machine other than the one you used for installation, open a browser and enter the following URL: `https://<connection-server-hostname>/admin/`. If you see a security warning, use the necessary UI controls to continue on. For example, for Firefox, you would click the **Advanced** button and then scroll down and click **Accept the risk and continue**. You see this warning because the default self-signed security certificate is being used.
11. Log in to Horizon Console using an account that belongs to the user or group account you specified when completing the Horizon Connection Server installation wizard.

For more information about installation and all the options, see the [Horizon Installation](#) guide.

Initial Configuration

The exercises in this chapter are about configuring the Connection Server so that you can create pools of VDI desktops and RDSH-published applications. In subsequent chapters, you will create and monitor desktop and application pools.

Some exercises in this chapter are mandatory, and some are optional. For example, the exercise [Create a Domain User Account and OUs in AD for Clone Operations](#) is optional in that you are not required to create a new domain user account and new Active Directory organizational units if you just want to set up a proof-of-concept (POC) environment.

Similarly, you are not required to set up the Events database. The Events database allows you to monitor logging operations in Horizon. If you do not complete the exercise [Create the Events Database](#), you can instead look directly in the log files if necessary, or you can configure logs to be sent to a Syslog server.

If you do not perform these optional exercises, configuring the Connection Server involves only three tasks: entering the license key, adding a vCenter Server, and designating an instant-clone domain administrator.

Create a Domain User Account and OUs in AD for Clone Operations

In this exercise, you perform the following preliminary tasks so that instant-clone desktops and RDSH servers can be automatically joined to a specified domain as they are created:

- Create a user account in Active Directory that has the required permissions for creating and deleting cloned desktops.
- Create one organizational unit (OU) in Active Directory for instant-clone desktops and another for instant-clone RDSH servers.

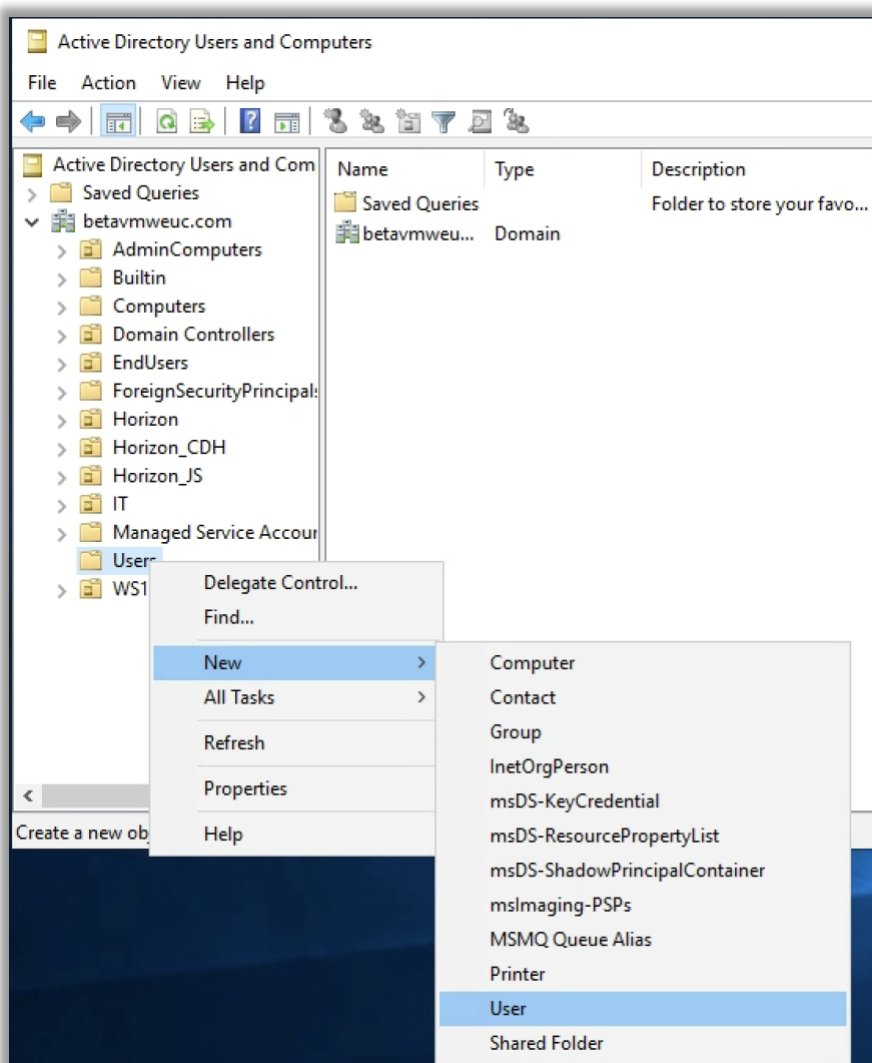
Note: This exercise shows how you would typically create an OU in a production environment and set the minimum required Active Directory domain privileges. However, for a test environment, you can skip this exercise and deploy the instant-clone virtual machines (VMs) to the Computers OU, and use a domain administrator account for the instant-clone domain.

Prerequisites for Creating OUs and the Domain Admin

To perform this exercise, you must have a user account for logging in to the domain controller as an administrator and creating users and OUs in Active Directory.

Create the Domain Admin User

1. On the Active Directory Domain Controller machine, log in as an administrator, and go to **Start button > Administrative Tools > Active Directory Users and Computers**.
2. Add a user, as follows: Expand the domain, right-click **Users**, select **New**, and select **User**.



3. Complete the New Object - User dialog box that appears.
For example, you could enter a logon name such as **Horizon-domain-user** into both the **Full name** text box and the **User logon name** text box, and click **Next**.
4. For the password information for this new user, use the following settings:
 - Enter a password of your choosing.
 - De-select **User must change password at next logon**, just for simplicity in this test environment.
 - Select **Password never expires**, just for simplicity.

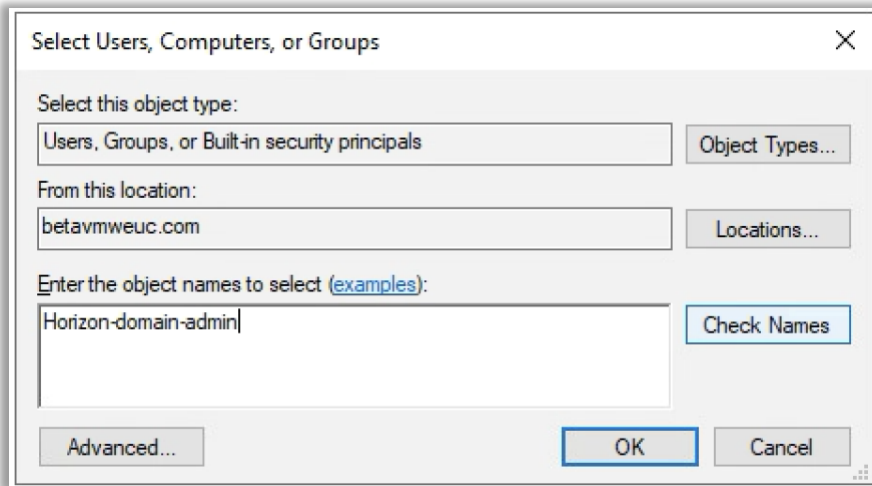
Now that you have a domain user account to use specifically for creating cloned VMs, you can add this user to the Active Directory OUs that will contain the VM computer accounts, as described in the next exercise. You will also assign permissions to this user so that the user account can create and delete VMs in the OUs.

Create OUs for Instant-Clone Desktops and RDSH Servers and Delegate Control

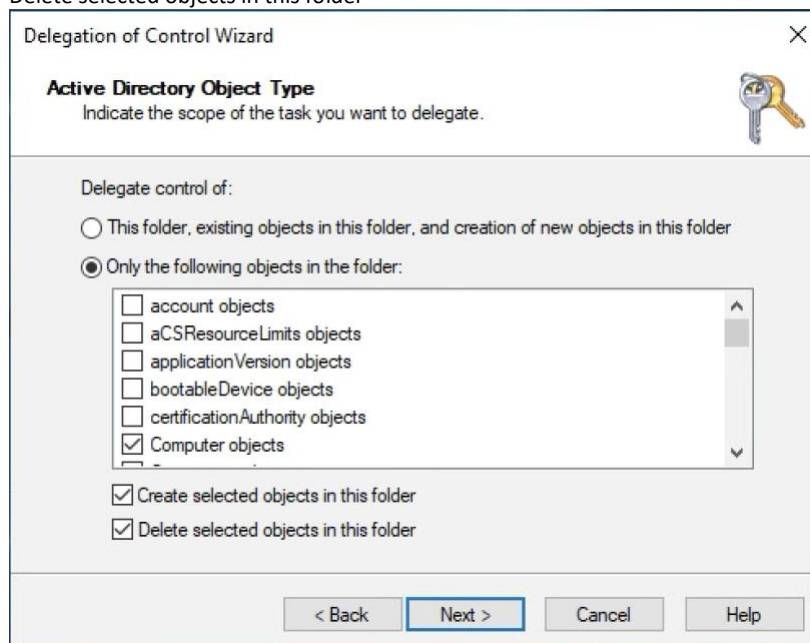
Although you could skip the first few steps of this procedure, which describe creating a new OU, and instead use the existing Computers OU in a test environment, you should still go through the rest of the procedure for whichever OU you use. That way, you can ensure that control of the OU is delegated to the correct user account, with the correct permissions granted on the correct AD objects.

1. On the Active Directory Domain Controller machine, log in as an administrator, and go to **Start button > Administrative Tools > Active Directory Users and Computers**.
2. Right-click the domain name, select **New**, and select **Organizational Unit**.
3. In the New Object - Organizational Unit dialog box that appears, enter a name, such as **Instant Clones**, and click **OK**. This OU is the Active Directory container in which the instant-clone computer accounts will be created. After you complete the text box, you can find the OU under the domain.
4. Right-click this OU you just created (which is the container) and select **Delegate Control**. The Delegation of Control wizard appears.
5. Click **Next** on the Welcome page, and click **Add** on the Users or Groups page.

6. Enter the name of the domain user you just created; for the example in this exercise, we use **Horizon-domain-user**, click **Check Names**, to verify that the name can be found in Active Directory, and click **OK**.

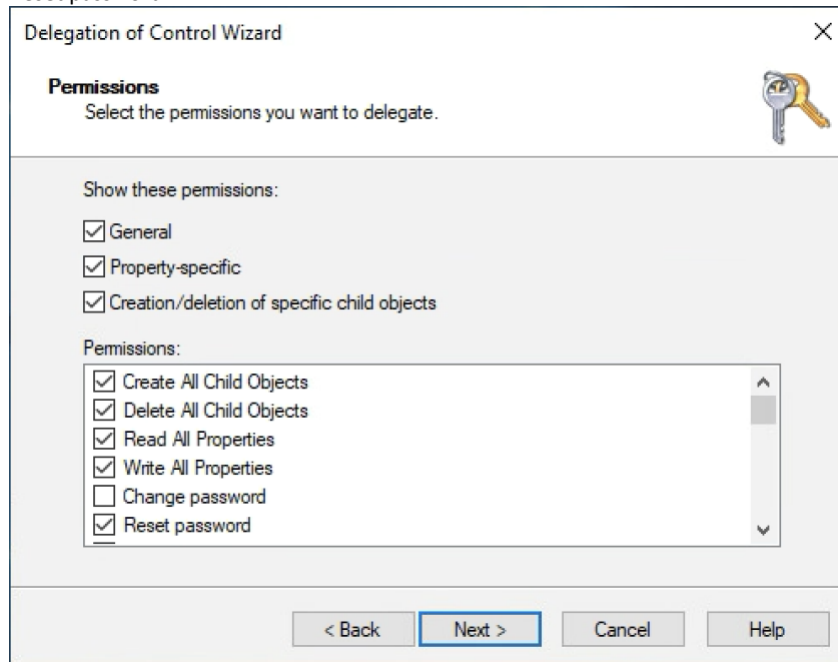


7. When you are returned to the Users or Groups page, click **Next**.
8. On the Tasks to Delegate page, select **Create a custom task to delegate**, and click **Next**.
9. On the Active Directory Object Type page, select the following check boxes before clicking **Next**:
- Computer objects
 - Create selected objects in this folder
 - Delete selected objects in this folder



10. On the Permissions page, select the following check boxes in the **Permissions** section before clicking **Next**:

- Create All Child Objects
- Delete All Child Objects
- Read All Properties
- Write All Properties
- Reset password



11. On the last page of the wizard, click **Finish**.

The user account now has the following complete list of required permissions, including permissions that are assigned by default:

- List Contents
 - Read All Properties
 - Write All Properties
 - Read Permissions
 - Reset password
 - Create Computer Objects
 - Delete Computer Objects
12. If you plan to perform the exercise for creating an instant-clone farm of RDSH servers, repeat this procedure to create an OU for the RDSH server computer accounts.
You might name the **OU RDSH Servers**.

Add the Product License Key

The first step of initial configuration after installing the Connection Server is to add a product license key. The first time you log in to the Connection Server, the Horizon Console opens to the Product Licensing and Usage page.

Important: Before you perform this exercise, you need a valid license. You can use an evaluation license if you have not yet purchased a license, which is listed on the [VMware Horizon Product Evaluation Center](#). For information about purchase options, see the [Compare section of the Horizon product page](#).

1. If you do not have the Horizon Console open, launch the Horizon Console by using one of the following methods:
 - If you are logged in to the server on which you installed the Connection Server, either double-click the **Horizon Administrator Console** desktop shortcut or open a browser and enter the following URL:
`https://localhost/admin/`.
 - If you are accessing the Horizon Console from a machine other than the one you used for installation, open a browser and enter the following URL: `https://<connection-server-hostname>/admin/`.

If you see a security warning, use the necessary UI controls to continue on. For example, for Firefox, you would click the **Advanced** button and then scroll down and click **Accept the risk and continue**. You see this warning because the default self-signed security certificate is being used.

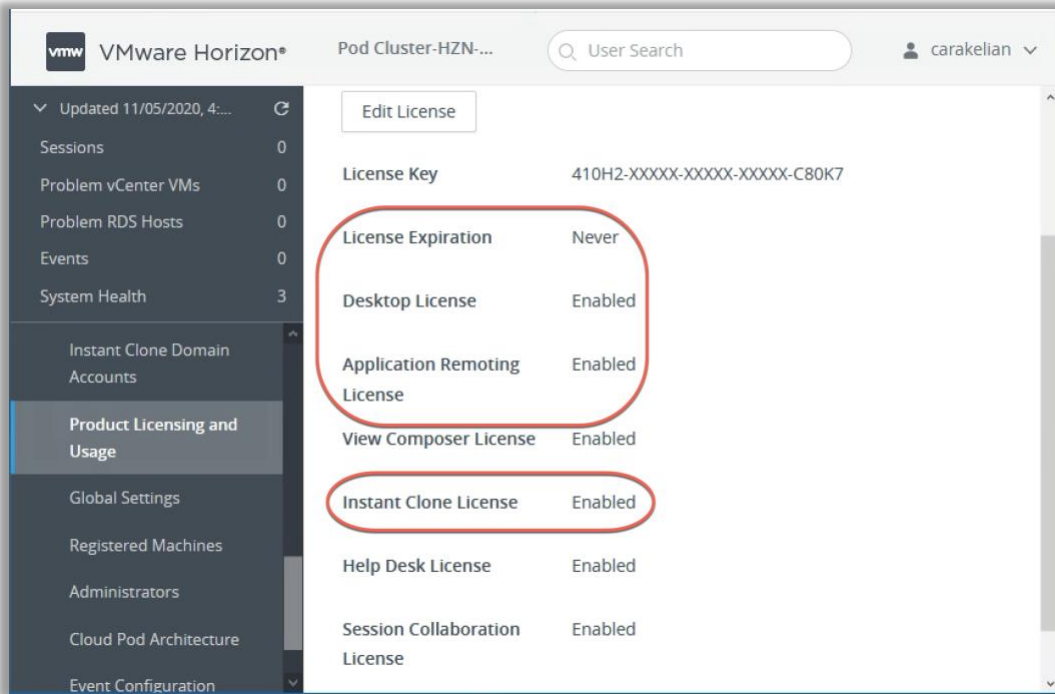
2. If the License and Usage page is not already displayed, navigate to **Settings > Product Licensing and Usage**, and then click the **Edit** button.
3. Enter the serial number of the product license key and click **OK**.

Important: For this tutorial, you are using a term license, rather than a subscription or universal subscription license. That is because this quick-start tutorial is designed to let you easily get a test environment up and running without having to purchase any Horizon software or contact any sales representative.

For production environments, VMware recommends purchasing and using a universal subscription license, which provides a single subscription license for all Horizon products, encompassing on-premises, Horizon subscription, and Horizon Cloud. For more information, see the [Horizon Universal License page](#).

Because this tutorial does not include using a universal subscription license, the tutorial does not walk you through the process of deploying a Horizon Cloud Connector and pairing it with a Connection Server so that the pod can be connected to the cloud-based Horizon Control Plane. For information about this task, see the topic [Enabling VMware Horizon for Subscription Licenses and Horizon Control Plane Services](#) in the *Horizon Installation* guide.

4. Verify that the license expiration date has not already passed and that the licenses for Desktop, Application Remoting, and Instant Clone are all enabled.



Add a vCenter Server Instance

Because the exercises in this tutorial walk you through the process of setting up a Horizon environment that uses a VMware vSphere infrastructure, the next task is to add a vCenter Server instance. vCenter Server creates and manages the virtual machines used in Horizon desktop pools. The Connection Server uses a secure channel (TLS/SSL) to connect to the vCenter Server instance.

Important: Before you perform this exercise, you need a vCenter Server user account that has the correct permissions. For more information, see [Configure a vCenter Server User for VMware Horizon](#) in the *Horizon Installation* guide.

1. In the Horizon Console, navigate to **Settings > Servers**, which takes you to the **vCenter Servers** tab, and click the **Add** button.
2. On the Add vCenter Server page, complete the following text boxes before clicking **Next**:
 - **Server address** – Enter the fully qualified domain name (FQDN) of the vCenter Server instance.
 - **User Name and Password** – Use the format `name@domain.com` for the name of the vCenter Server user account.
 - You can leave the default settings for the other text boxes.

3. If an Invalid Certificate Detected prompt is displayed:
 - a. Click **View Certificate**.
 - b. In the Certificate Information window that appears, review the thumbprint of the default self-signed certificate that was generated during installation, and click **Accept**.
4. Click **Next** on the rest of the wizard pages to accept the defaults, and on the Ready to Complete page, click **Submit**.

Step	Option	Value
4 Ready to Complete	Max Provision	20
	Max Power	50
	Max View Composer Operations	12
	Max View Composer Provision	8
	Max Instant Clone Engine Provision	20
	View Composer State	Do not use View Composer
	Enable View Storage Accelerator	Yes
	Default host cache size (MB)	1,024
	VM Disk Space Reclamation	Yes
	Deployment Type	General

Buttons: Cancel, Previous, Submit

You are returned to the **Servers > vCenter Servers** tab, and the server you just added appears in the list.

Add an Instant-Clone Domain Administrator

You use the Horizon Console to specify the user account for joining instant-clone VMs to the Active Directory domain.

Important: Before you perform this exercise, you must have a domain user account that has the required Active Directory permissions so that cloned VMs can be joined to the domain. These include permissions to create and delete computer objects, and to write properties in the domain or in the OUs (organizational units) that you select when creating desktop pools or server farms in later exercises. You have already created this user account if you performed the exercises [Create the Domain Admin User](#) and [Create OUs for Instant-Clone Desktops and RDSH Servers and Delegate Control](#).

Tip: In a test environment, you could use an account that is a member of the Domain Administrators group, which has all the required privileges.

1. In the Horizon Console, navigate to **Settings > Instant Clone Domain Accounts**, and click the **Add** button.
2. On the Add Domain Admin page that appears, select the domain from the list, enter the user name and password for the user you created in the exercise [Create the Domain Admin User](#), and click **OK**.

You are returned to the **Instant-Clone Engine Domain Accounts** page, and the user account you just added appears in the list.

Create and Configure the Events Database

In this exercise, you create an Events database to log Horizon events to a SQL Server instance, making the event data available to analytics software. For example, you can find the following types of events in the database:

- Alerts that report system failures and errors
- End-user actions, such as logging and starting desktop and application sessions
- Administrator actions, such as adding entitlements and creating desktop and application pools
- Statistical sampling, such as recording the maximum number of users over a 24-hour period

For details about the types of information recorded, see [Integrating VMware Horizon with the Event Database](#) in the *Horizon Administration* guide. The Events database is not required for every Horizon environment. Alternatively, or in addition to using the Events database, you can configure the Connection Server to send events to a Syslog server or create a flat file of events written in Syslog format. See [Configure Event Logging to File or Syslog Server in Horizon Console](#) in the *Horizon Installation* guide.

Prerequisites for Setting Up the Events Database

To perform this exercise, you need the following:

- **SQL Server instance** – This is the database server on which you will create the Events database. For the example in this exercise, we used Microsoft SQL Server 2019.
For the most up-to-date information about supported databases, see the VMware Product Interoperability Matrixes at http://www.vmware.com/resources/compatibility/sim/interop_matrix.php. For **Solution/Database Interoperability**, after you select the product and version, for the **Add Database** step, to see a list of all supported databases, select **Any** and click **Add**.
- **Microsoft SQL Server Management Studio** – For the example in this exercise, we used Microsoft SQL Server Management Studio 18. The instructions might differ slightly for different versions of SQL Server Management Studio.
- **Microsoft SQL Server Configuration Manager** – For the example in this exercise, we used SQL Server 2019 Configuration Manager. The instructions might differ slightly for different versions of SQL Server Configuration Manager.
- **SA credentials** – To create the necessary logins for the database, you will log in to the SQL Server instance as the sysadmin (SA) or as a user account with SA privileges.

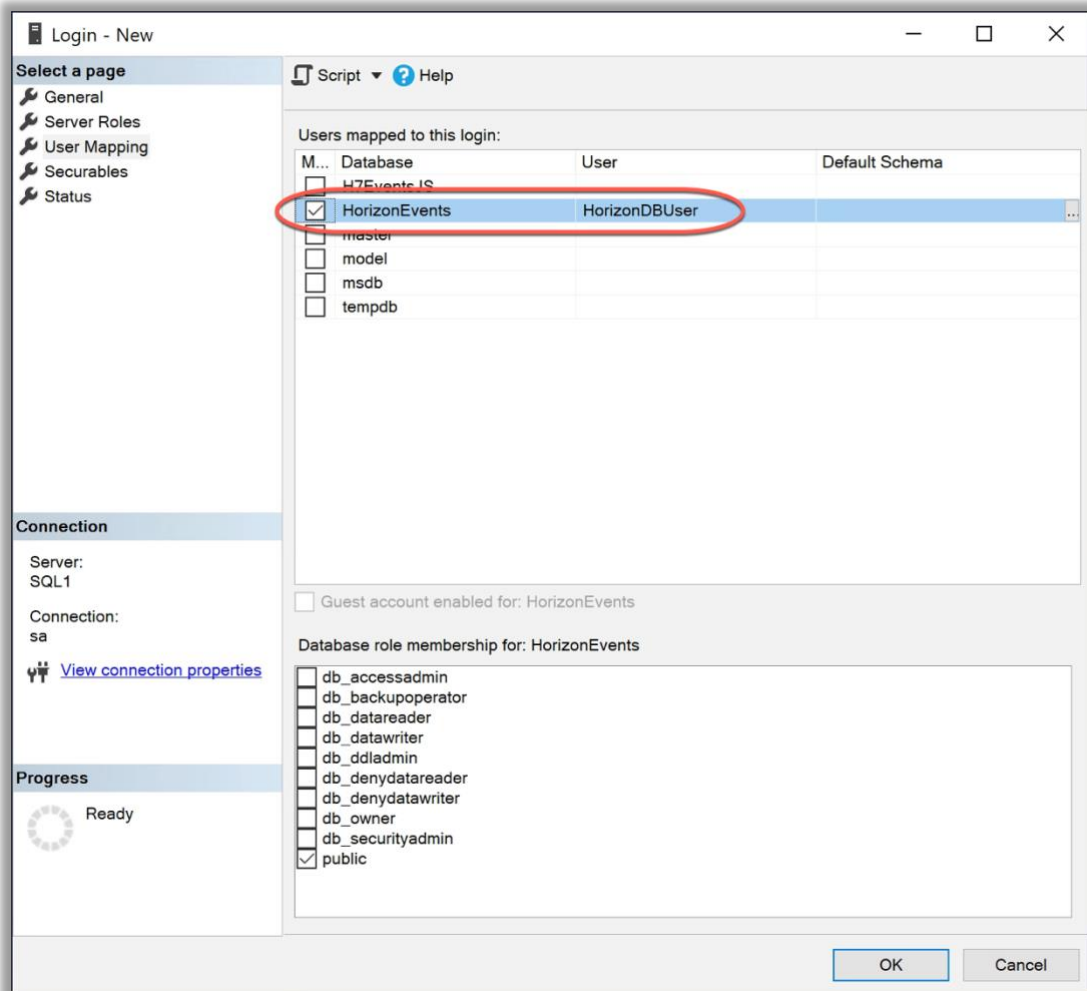
Create the Events Database and Its Login User

1. On the VM where SQL Server and SQL Server Management Studio are installed, start SQL Server Management Studio.
2. Complete the Connect to Server dialog box as follows before clicking **Connect**:
 - **Server type** – Select **Database Engine**.
 - **Server name** – If the server name does not appear by default, select it.
 - **Authentication** – Select **SQL Server Authentication**.
 - **Login and Password** – Use the SA account credentials.
3. In the Object Explorer, right-click **Databases**, and select **New Database** from the submenu.
4. Name the database and click **OK**.
For this example, the database name is **HorizonEvents**. This database now appears in the list when you expand the **Databases** folder in the Object Explorer.
5. To create a login so that the Connection Server can access the database to log events, expand the **Security** folder, right-click **Logins**, and select **New Login**.
6. Complete the General Settings page, as follows:

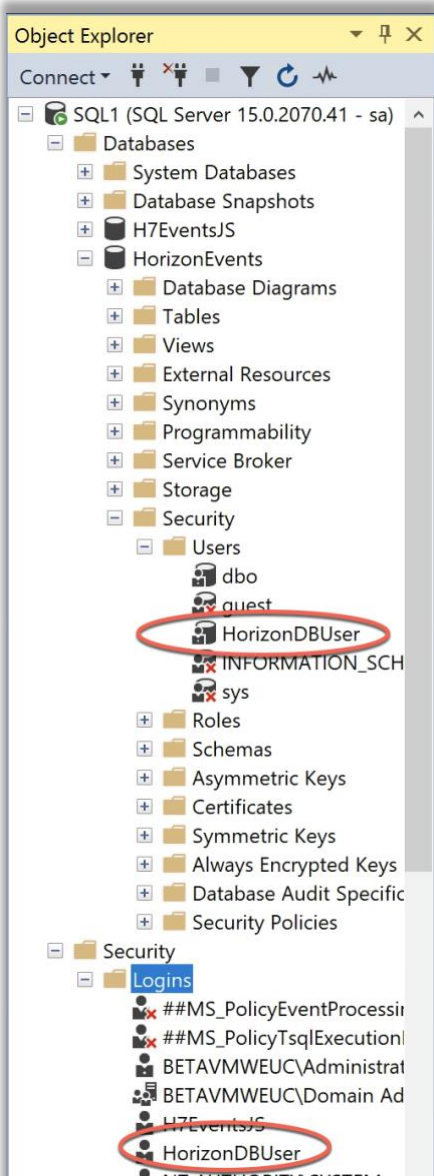
- a. Enter a login name to use for the Connection Server machine, using ASCII characters only; for example, **HorizonDBUser**.
- b. Select **SQL Server authentication**, and create a password.
- c. De-select **Enforce password policy**. For the purposes of this exercise, you do not need to use password policies.
- d. Either leave **master** as the default database or select the **HorizonEvents** database as the default database.
- e. Do not click **OK** yet because you must fill out the other pages.

7. From the **Select a Page** list on the left, select **Server Roles**, and select the **sysadmin** check box on that page.

8. From the **Select a Page** list on the left, select **User Mapping**, and in the **Users mapped to this login** box, select the check box for the **HorizonEvents** database, and click **OK**.

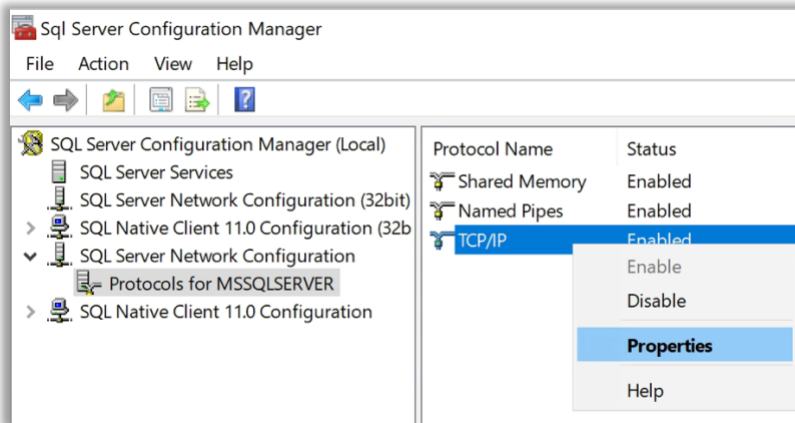


The login user is added in the Object Explorer pane, under the **Security > Logins** folder, and under the **Databases > HorizonEvents > Security > Users** folder.

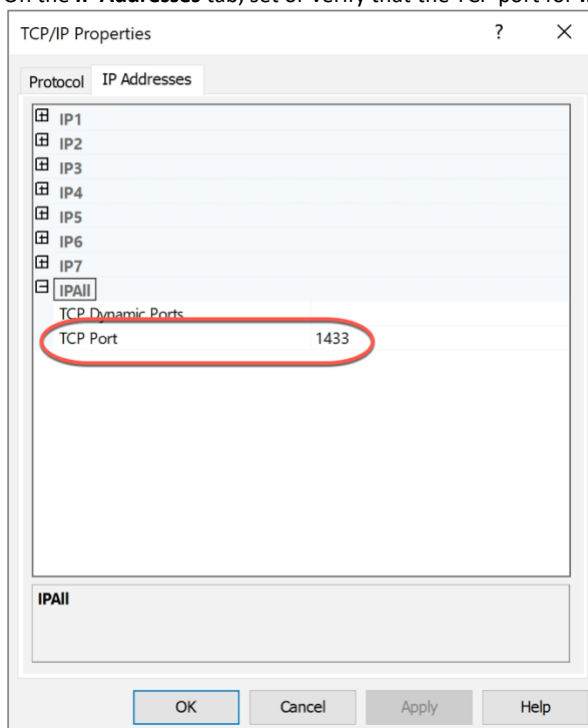


Configure TCP/IP Properties for the Database Server

1. On the VM where SQL Server and SQL Server Configuration Manager are installed, start SQL Server Configuration Manager.
2. Expand **SQL Server Network Configuration**, select **Protocols for <server name>**, and in the list of protocols, right-click **TCP/IP**, and select **Properties**.



3. In the TCP/IP Properties window that appears, on the **Protocol** tab, set or verify that the **Enabled** property is set to **Yes**, and then click the **IP Addresses** tab.
4. On the **IP Addresses** tab, set or verify that the TCP port for **IPAll** is set to the default port 1433, and click **OK**.



Configure the Events Database in the Horizon Console

1. In the Horizon Console, navigate to **Settings > Event Configuration**.
2. In the **Event Database** section, click the **Edit** button.
3. Complete the Edit Event Database page, as follows, before clicking **OK**:
 - **Database Server** – Enter the FQDN.
 - **Database Type** – Accept the default, **Microsoft SQL Server**.
 - **Port** – Accept the default port number (**1433**) used to access the database server.
 - **Database Name** – Enter the event database name created on the database server; for this example, **HorizonEvents**.
 - **User name and Password** – Enter the credentials for the user you created for this database in [Create the Events Database and Its Login User](#). For this example, the user name is **HorizonDBUser**.
 - **Table Prefix** – Enter **VE_** (for View Events).

The screenshot shows the 'Edit Event Database' dialog box with the following fields and values:

- Database Server:** sql1.betavmware.com
- Database Type:** Microsoft SQL Server
- Port:** 1433
- Database Name:** HorizonEvents
- User Name:** HorizonDBUser
- Password:** (masked with dots)
- Confirm Password:** (masked with dots)
- Table Prefix:** VE_

At the bottom right, there are 'Cancel' and 'OK' buttons.

The configuration settings you entered are displayed on the Event Configuration page.

4. To verify that the connection to the database is working, in the Horizon Console, navigate to **Monitor > Events**, and verify that some log entries appear in the table.

Creating Single-User Desktop Pools

With single-user desktops, each virtual machine allows a single end-user connection at a time. In contrast, with session-based desktops, one RDSH server can accommodate many concurrent user connections. In this chapter, you will create an instant-clone single-user desktop pool.

A shared-session, RDSH desktop pool has different characteristics than a single-user automated desktop pool. Creating an RDSH desktop pool is beyond the scope of this quick-start tutorial, but if you are interested in this topic, see [Create a Published Desktop Pool](#) in the *Setting Up Published Desktops and Applications in Horizon* guide. In a later chapter of this tutorial, you will complete an exercise for creating an RDSH server farm and a published application pool.

Besides Windows-based desktops, you can create Linux-based desktops. For more information, see the document [Setting Up Linux Desktops in Horizon](#).

Deploy an Instant-Clone Desktop Pool

A clone is a copy of a golden VM image, with a unique identity of its own, including a MAC address, UUID, and other system information. VMware instant-clone provisioning improves on and accelerates the process of creating cloned VMs over the previous Horizon Composer [linked-clone technology](#). In addition, instant clones require less storage and less expense to manage and update because the desktop is deleted when the user logs out, and a new desktop is created using the latest golden image.

Creating an instant-clone desktop pool or RDSH server farm is a two-part process:

- Publishing, also called priming, the golden VM image
- Provisioning the VMs in the pool or farm

Publishing the golden image can take from 7 to 40 minutes, depending on the type of storage you are using. Provisioning the VMs takes only 1 or 2 seconds per VM. You can perform these tasks at separate times, so that the provisioning process occurs either at a scheduled time or immediately after the publishing process is complete.

The Add Desktop Pool wizard or the Add Farm wizard in the Horizon Console guides you through the process of publishing the golden image. Completing the wizard for instant clones is similar to adding any type of pool or farm.

Important: If your session in the Horizon Console is idle for more than a few minutes, you might be automatically logged out, and if you are in the middle of creating a desktop pool, your changes will be lost. You can increase the session timeout interval by navigating to **Settings > Global Settings** and editing the **View Administrator Session Timeout** setting.

Prerequisites for Deploying an Instant-Clone Desktop Pool

To perform this exercise, you need the following:

- **Golden VM** – Before you can deploy a pool of desktops, you must create an optimized golden VM image, which includes installing and configuring a Windows operating system in a VM, optimizing the OS, and installing the various VMware agents required for desktop pool deployment. For step-by-step instructions, see the guide [Creating an Optimized Windows Image for a VMware Horizon Virtual Desktop](#).
- **VM snapshot** – After you create and configure the golden VM, use vSphere Web Client to power it off and take a VM snapshot.
- **Connection Server** – For installation and setup instructions, see the exercises [Install Horizon the Connection Server](#), [Add the Product License Key](#), and [Add a vCenter Server Instance](#).
- **Instant-clone domain administrator** – You must have created and added an instant-clone domain administrator, as described in the exercises [Create the Domain Admin User](#) and [Add an Instant-Clone Domain Administrator](#).
- **AD OU** – You must have determined which Active Directory OU to use for storing instant-clone computer accounts. In a test environment, you can use the Computers OU. In a production environment, VMware recommends that you create a specific OU and domain user. But you must also delegate the minimum required permissions, as described in the exercise [Create OUs for Instant-Clone Desktops and RDSH Servers and Delegate Control](#).
- **VM folder** – (Optional) Having a specific VM folder in the vCenter Server inventory helps you locate and manage the virtual desktops in the instant-clone pool.

Run the Add Pool Wizard to Deploy an Instant-Clone Desktop Pool

1. In the Horizon Console, navigate to **Inventory > Desktops**, and click the **Add** button.
Note: The format of the URL for accessing the web-based console is:
`https://<connection-server-FQDN>/admin`.
2. In the Add Pool wizard that opens, with **Type** selected in the left pane, select **Automated Desktop Pool**, and click **Next**.
3. On the vCenter Server page, select **Instant Clone**, and click **Next**.
The vCenter Server you added during initial configuration should appear and be selected in the vCenter Server list.
4. On the User Assignment page, select **Floating**, and click **Next**.
Instant-clone pools can use either floating or dedicated user assignment. For this exercise, we use floating assignment.
 - **Dedicated assignment** – Each desktop is assigned to a specific user. A user logging in for the first time gets a desktop that is not assigned to another user. The user always gets this same desktop after logging in, and this desktop is not available to any other user.
 - **Floating assignment** – Users get a random desktop every time they log in. When a user logs out, the desktop is deleted. With automatic deletion, you keep only as many VMs as you need at one time.

5. On the Storage Optimization page, select **Use separate datastores for replica and OS disks**, and click **Next**.
For this exercise, if possible, use separate datastores so that you can see the extra settings in the next window. With separate datastores, you can place the replica VM on a solid-state, disk-backed datastore. Solid-state disks have low storage capacity but high read performance, typically supporting 20,000 IOPS. Separate datastores are used in tiered-storage models.
Note: For information about VMware Virtual SAN, see the [VMware vSAN product page](#).
6. On the Desktop Pool ID page, complete the settings, as follows, before clicking **Next**:
 - **ID** – For this exercise, use the pool ID **Win-10-instant-clone**.
 - **Display Name** – For this exercise, use **Windows 10 Desktop**.
Users will see this display name when they log in using Horizon Client or the HTML Access web client. If you do not provide a display name, the pool ID is used for the display name.
 - **Access Group** – If you do not specify an access group, the pool is placed in the root access group. For more information about access groups, see the product documentation topic [Manage and Review Access Groups](#) in the *Horizon Administration* guide.
7. On the Provisioning Settings page, complete the settings, as follows, before clicking **Next**:
 - **Naming Pattern** – For this exercise, you use **Win10-IC**. This naming pattern helps you identify Windows 10 instant clones in Horizon Console.
 - **Provision Machines** – Select **Machines on Demand**, and use the default minimum of **1**.
 - **Desktop Pool Sizing** – Set **Maximum Machines** to **10** (for the purposes of this exercise), and set **Spare (Powered On) Machines** to **1**.
In a production environment, instant-clone pools have been tested to support up to 4,000 desktops (2,000 is recommended, however).

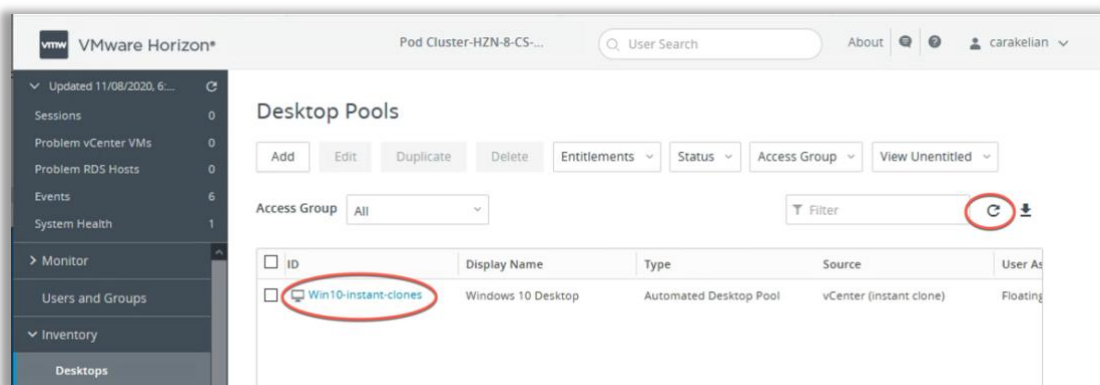
- Use the defaults for the other settings.

8. On the vCenter Settings page, complete the **Default Image** settings, as follows:
 - a. For the **Parent VM in vCenter** setting, click **Browse** to select the golden Windows 10 VM you created according to the instructions in [Creating an Optimized Windows Image for a VMware Horizon Virtual Desktop](#), and click **Submit**.
 - b. For the **Snapshot** setting, click **Browse** to select the snapshot you created as part of the prerequisites for this exercise, and click **Submit**.
The snapshot will be used as the default image for creating the pool. For instructions on creating the snapshot, see the guide [Creating an Optimized Windows Image for a VMware Horizon Virtual Desktop](#).
9. In the **Virtual Machine Location** section, click **Browse** to select a VM folder if you created one as described in the prerequisites for this exercise.
10. In the **Resource Settings** section, click **Browse** to select the appropriate vCenter resource for each setting. As with the other settings on this page, to complete each setting, you will select from the resources that are already set up in your vCenter Server and that are specific to your environment.
Important: For the **Network** setting, leave the default, which means the **Use network from current parent VM image** check box is selected.

11. With the vCenter Settings page completed, click **Next**.

12. On the Desktop Pool Settings page, leave the defaults and click **Next**.
Note: To learn more about any of these settings, see the topic [Worksheet for Creating an Instant-Clone Desktop Pool](#), in the *Setting Up Virtual Desktops in Horizon* guide.
13. On the Remote Display Settings page, select the **Allow Session Collaboration** check box, and click **Next**.
14. Complete the Guest Customization page, as follows:
- Verify that the correct domain and domain admin account are selected. This is the account you created in the exercise [Add an Instant-Clone Domain Administrator](#).
 - For **AD container**, click **Browse** and select the OU that you created in the exercise [Create OUs for Instant-Clone Desktops and RDSH Servers and Delegate Control](#).
 - Leave the other default settings, and click **Next**.
15. On the Ready to Complete page, click **Submit**.
 You are returned to the **Inventory > Desktops** page, called Desktop Pools.

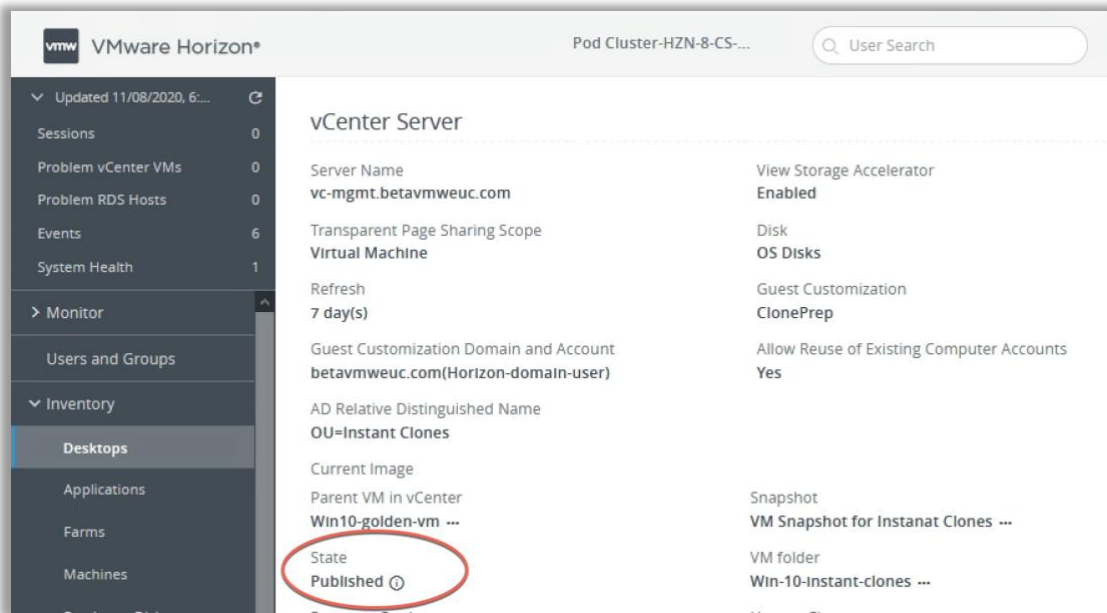
16. To monitor pool creation, click the name of the desktop pool in the list.
If you do not see the name in the list, click the Refresh icon above the table.



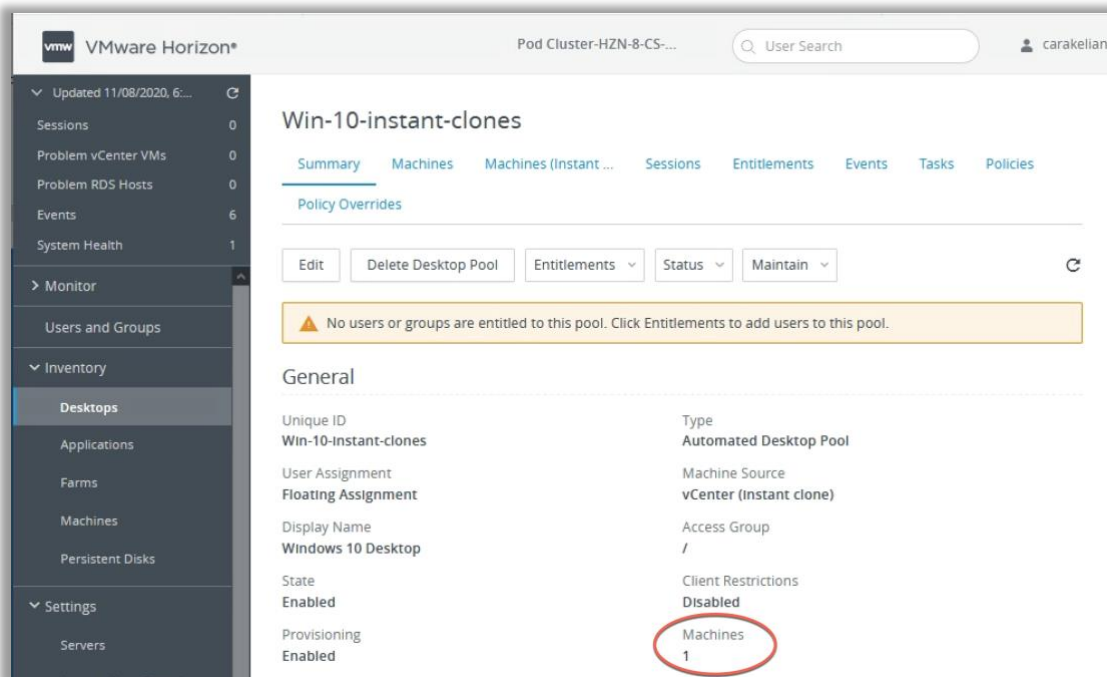
Because the VM image must be published before the instant clones can be created, the process might take several minutes.

17. On the pool details page for the instant-clone pool, scroll down to the **vCenter Server** section, and note the **State** field and the **Pending Image** field.

The **Pending Image** field shows the progress of the publish operation. When the pool is finished being created, the **State** field shows **Published**.



When the instant clone is finished being created, if you scroll up to the **General** section, the **Machines** field shows 1.



Note: The **Summary** tab is shown in this screenshot. To see details of the particular instant clone you just created, click the **Machines** tab or the **Machines (Instant-Clone Details)** tab, and for even more details, click the machine name in the list.

The screenshot displays the VMware Horizon management interface. On the left is a navigation pane with categories like 'Updated', 'Monitor', 'Inventory', and 'Settings'. The 'Machines' item under 'Inventory' is selected. The main area shows the details for a machine named 'Win-10-clone1'. At the top, there are tabs for 'Summary' (active), 'vCenter Settings', and 'Events'. Below the tabs are four action buttons: 'Restart Desktop', 'Reset Virtual Machine', 'Recover', and 'Remove'. The 'General' section lists properties such as Name, Access Group, Desktop Pool, Agent Version, Assigned User, DNS Name, Status, and Connected User. The 'Image Settings' section lists Base Image, Last Compose, Pending Image, and Task.

General	
Name	Win-10-clone1
Assigned User	N/A
Access Group	/
DNS Name	win-10-clone1.betavmweuc.com
Desktop Pool	Win-10-Instant-clones
Status	Available
Agent Version	8.0.0
Connected User	-

Image Settings	
Base Image	/DC/vm/Win10-golden-vm/VM Snapshot for Instant Clones
Pending Image	-
Last Compose	11/08/2020, 8:23 PM
Task	None
Datastores	XT01-BETA-MGMT-1

Push a New Image to an Instant-Clone Desktop Pool

To manage OS patches and software updates with instant clones, you use the push-image operation. Because the provisioning of instant clones takes so little time, it is not necessary to plan for maintenance windows.

When a user logs out of their desktop, the desktop is deleted and recreated. This approach to desktop deletion and recreation staggers the patching operation across desktops, eliminates boot storms, reduces storage IOPS, and creates less of a load on the vCenter Server.

To perform this exercise, you must satisfy the following prerequisites:

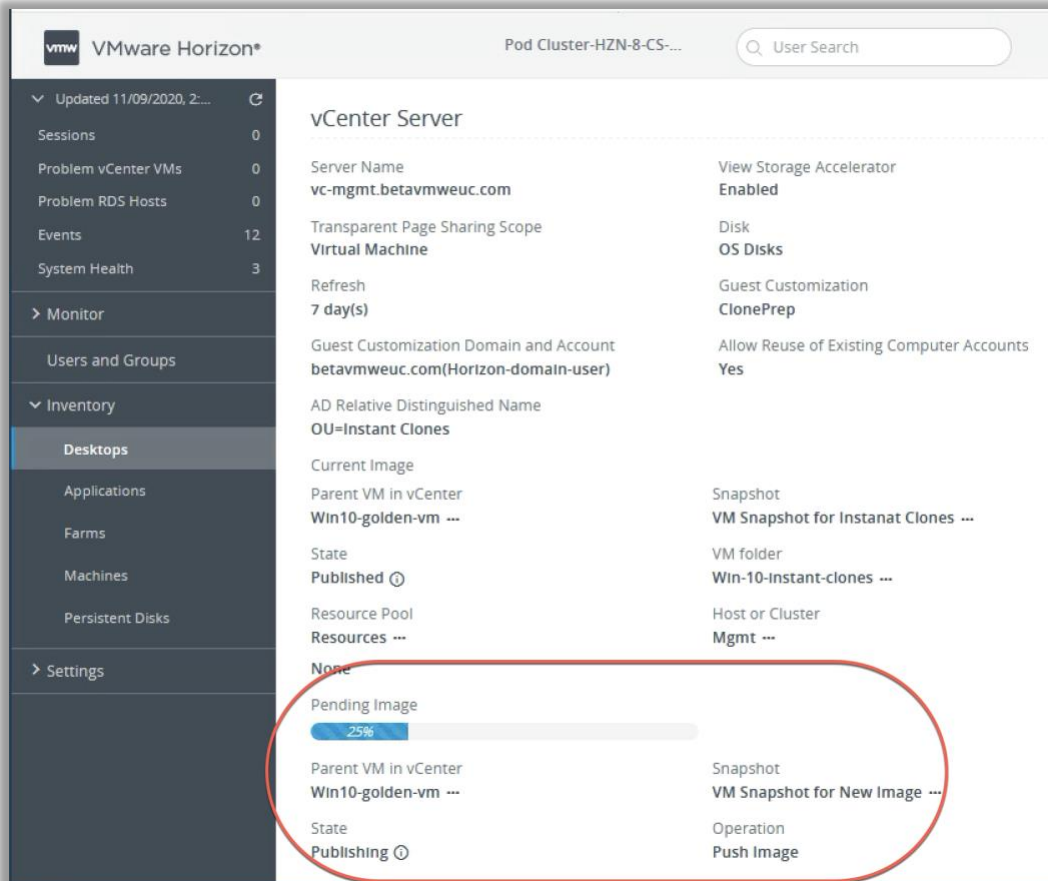
- **Instant-clone desktop pool** – You must have completed the exercise [Deploy an Instant-Clone Desktop Pool](#).
- **New VM snapshot** – You must have a new image to push to the desktop pool. Therefore, use vSphere Web Client, select the VM that you created for deploying the instant-clone pool, and create a new VM snapshot. For details, see the vSphere documentation topic [Take a Snapshot of a Virtual Machine](#).

To push a new image to a desktop pool:

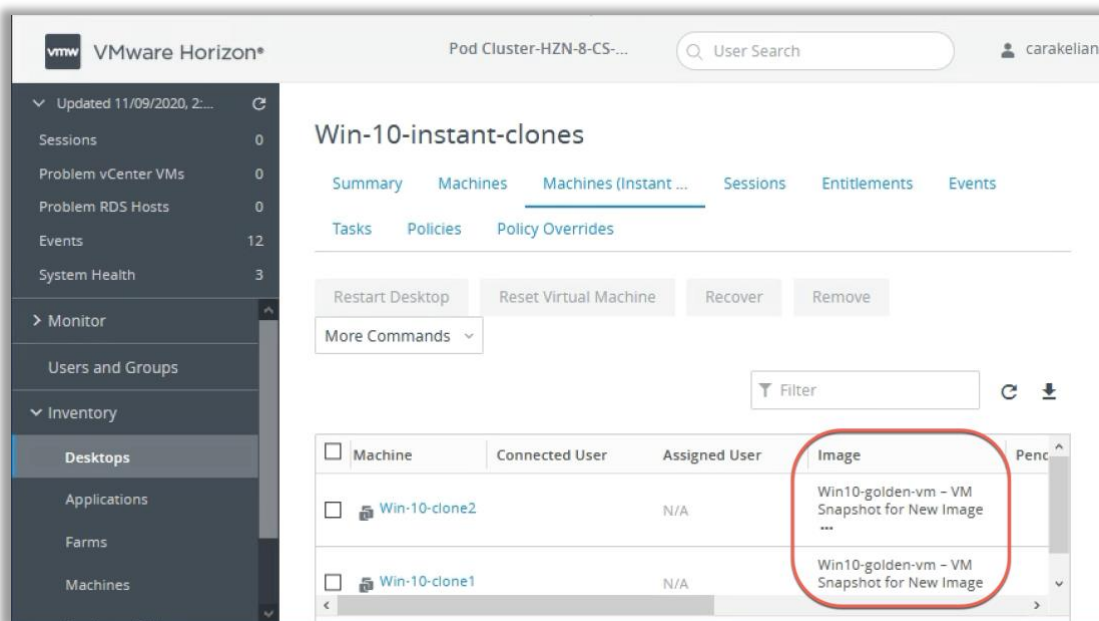
1. In the Horizon Console, navigate to **Inventory > Desktops** and click the name of the desktop pool in the list. The **Summary** tab for that desktop pool appears.
2. Click the **Maintain** button, and select **Schedule** from the drop-down list. The Schedule Push Image wizard appears.
3. On the Image page, select the new snapshot you created, as mentioned in the prerequisites, and click **Next**. For this exercise, we select a new snapshot taken of the same golden VM, but you can also use this page to navigate to a different VM and one of its snapshots.
4. On the Schedule page, leave the defaults, and click **Next**. Because you are using the default start time, the push starts after you complete the wizard.
Note: The default is **Wait for users to log off**. If, instead, you select to force users to log off, you can give users a warning and a grace period of 5 minutes, by default. To edit this setting, navigate to **Settings > Global Settings**, and click **Edit** on the **General Settings** tab.

- On the Ready to Complete page, click **Finish**.

You are returned to the **Summary** tab for the desktop pool, where the pending image for the push operation is displayed in the **vCenter Server** section. You can watch the progress under the **Pending Image** field. The state changes from **Publishing** to **Published**, and then the current image will be unpublished.



6. Click the **Machines (InstantClone Details)** tab to monitor which desktops are using which image.



The screenshot displays the VMware Horizon console interface. The left sidebar shows the navigation menu with 'Desktops' selected under the 'Inventory' section. The main content area is titled 'Win-10-instant-clones' and shows the 'Machines (InstantClone Details)' tab. Below the tabs, there are buttons for 'Restart Desktop', 'Reset Virtual Machine', 'Recover', and 'Remove'. A 'More Commands' dropdown is also visible. A table lists the clones, with the 'Image' column circled in red. The table has columns for 'Machine', 'Connected User', 'Assigned User', 'Image', and 'Penc'.

Machine	Connected User	Assigned User	Image	Penc
☐ Win-10-clone2		N/A	Win10-golden-vm - VM Snapshot for New Image ...	
☐ Win-10-clone1		N/A	Win10-golden-vm - VM Snapshot for New Image ...	

Creating RDSH-Published Desktops and Applications

VMware Horizon published desktops and applications are based on sessions to RDSH servers. That is, administrators use Microsoft Remote Desktop Services (RDS) to provide users with desktop and application sessions on RDS hosts. Delivering published applications and desktops is a very simple process:

1. Create an RDSH server farm from a golden VM image, which automatically clones the number of servers you specify.
2. Publish one or multiple application pools with one trip through the Add Application Pool wizard.
3. Publish a desktop pool so that multiple users can access session-based shared desktops from RDSH servers. This tutorial does not include an exercise for publishing a desktop pool. Instead, see [Creating Published Desktop Pools](#), in the guide *Setting Up Published Desktops and Applications in Horizon*.
Note: It is also possible to enable Windows Server machines to be used as single-user desktops rather than RDSH session-based shared desktops. For information, see the topic [Prepare Windows Server Operating Systems for Desktop Use](#) in the guide *Setting Up Virtual Desktops in Horizon*.
4. Perform image maintenance tasks for RDSH servers.
 This tutorial does not include an exercise for refreshing RDSH server VMs from a golden image virtual machine. Instead, see [Schedule Maintenance for an Automated Instant-Clone Farm in Horizon](#), in the guide *Setting Up Published Desktops and Applications in Horizon*.

The exercises that follow walk you through creating an instant-clone server farm and publishing applications. On client devices, a published application looks and behaves like it would if it were a locally installed application, providing seamless integration into the user experience of the client operating system. For example, on Windows client devices, the application icon for the published application appears in the taskbar just as it would for a locally installed application.

Important: This chapter describes publishing applications that are hosted on RDSH servers. With Horizon, you can also publish applications that are hosted on virtual desktops, such as a Windows 10 desktop pool. This feature is especially important if you want to publish applications that are certified only on Windows 10 or that require drivers that are not supported on RDSH servers. For more information, see [Deploying Applications that Run on Desktop Pools with VM Hosted Applications](#) in the guide *Setting Up Virtual Desktops in Horizon*.

Create an Instant-Clone RDSH Server Farm

A farm can contain from 1 to 500 RDSH servers. For the exercises in this guide, you create an automated farm of RDSH servers, which is similar to creating an automated pool of instant-clone desktops. With this feature, you do not need to create and configure each RDSH server separately.

Prerequisites for Creating an Instant-Clone Server Farm

To perform this exercise, you need the following:

- Golden VM** – Before you can deploy a farm of RDSH servers, you must create an optimized golden image, which includes installing and configuring a Windows operating system in a VM, optimizing the OS, and installing the various VMware agents required for server farm deployment. For step-by-step instructions, see the guide [Creating an Optimized Windows Image for a VMware Horizon Virtual Desktop](#). After you create and configure the VM, power it off and take a VM snapshot.
Important: The golden VM for RDSH servers must have the appropriate RDSH roles and services installed. See the topics [Prepare an RDS Host Golden Image Virtual Machine](#) and [Prepare Windows Server Operating Systems for Remote Desktop Services \(RDS\) Host Use](#) in the guide *Setting Up Published Desktops and Applications in Horizon*.
- VM snapshot** –After you create and configure the golden VM, use vSphere Web Client to power it off and take a VM snapshot.
- AD OU** – You must have determined which Active Directory OU to use for storing instant-clone computer accounts. In a test environment, you can use the Computers OU. In a production environment, VMware recommends that you create a specific OU and domain user. You must also delegate the minimum required permissions, as described in the exercise [Create OUs for Instant-Clone Desktops and RDSH Servers and Delegate Control](#).
Note: For the server farm OU, give the OU a descriptive name such as RDSH Servers.
- Instant-clone domain administrator** – You must have created and added an instant-clone domain administrator, as described in the exercises [Create the Domain Admin User](#) and [Add an Instant-Clone Domain Administrator](#).
- VM folder** – (Optional) If desired, create VM folder in the vCenter Server inventory for the farm. Having a specific folder helps you locate and manage the RDSH servers in the instant-clone farm.
- Applications** – The applications you provide to end users can be either installed directly on the RDSH server or dynamically attached as VMware App Volumes AppStacks (If you are using App Volumes 2.x) or App Volumes packages (if you are using App Volumes 4). Before you begin this exercise, install any applications that you want to have in the base image, available for all users.
Note: To install applications directly on an RDSH server, place the host into RD-Install mode, install the desired applications, and place the host back into RD-Execute mode. For more information, see the Microsoft TechNet article [Learn How To Install Applications on an RD Session Host Server](#). If you plan to use App Volumes AppStacks or packages, be sure to install the App Volumes Agent, as described in the guide [Creating an Optimized Windows Image for a VMware Horizon Virtual Desktop](#).

Run the Add Farm Wizard to Deploy an Instant-Clone RDSH Server Farm

1. In the Horizon Console, navigate to **Inventory > Farms**, and click the **Add** button.
Note: The format of the URL for accessing the web-based console is:
<https://<connection-server-FQDN>/admin>.
2. In the Add Farm wizard that opens, with **Type** selected in the left pane, select **Automated Farm**, and click **Next**.
3. On the vCenter Server page, select **Instant Clone**, and click **Next**.
 The vCenter Server you added during initial configuration should appear and be selected in the vCenter Server list.
4. On the Storage Optimization page, click **Next**.
Note: Although you are not using Virtual SAN for this exercise, if you would like information about VMware Virtual SAN, see the [VMware vSAN product page](#).
5. On the Identification and Settings page, complete the settings, as follows, before clicking **Next**:
 - **ID** – For this exercise, use the ID **RDSH Farm**.
 - **Access Group** – If you do not specify an access group, the pool is placed in the root access group. For more information about access groups, see the product documentation topic [Manage and Review Access Groups](#) in the *Horizon Administration* guide.
 - Leave the defaults for the other settings.
6. On the Load Balancing and Settings page, click **Next**.
 For more information about these settings, see [Load Balancing Settings](#) in the guide *Setting Up Published Desktops and Applications in Horizon*.
7. On the Provisioning Settings page, complete the settings, as follows, before clicking **Next**:
 - **Naming Pattern** – For this exercise, you use **RDSH-**. This naming pattern helps you identify RDSH server instant clones in Horizon Console.
 - **Farm Sizing** – Set **Maximum Machines** to **10** (for the purposes of this exercise), and set **Minimum Number of Ready (Provisioned) Machines** to **1**.
8. On the vCenter Settings page, complete the **Default Image** settings, as follows:
 - a. For the **Parent VM in vCenter** setting, click **Browse** to select the golden RDSH server VM you created according to the instructions in [Creating an Optimized Windows Image for a VMware Horizon Virtual Desktop](#), and click **Submit**.
 - b. For the **Snapshot** setting, click **Browse** to select the snapshot you created as part of the prerequisites for this exercise, and click **Submit**.
 The snapshot will be used as the default image for creating the farm. For instructions on creating the snapshot, see the guide [Creating an Optimized Windows Image for a VMware Horizon Virtual Desktop](#).
9. In the **Virtual Machine Location** section, click **Browse** to select a VM folder if you created one as described in the prerequisites for this exercise.

10. In the **Resource Settings** section, click **Browse** to select the appropriate vCenter resource for each setting.
As with the other settings on this page, to complete each setting, you will select from the resources that are already set up in your vCenter Server and that are specific to your environment.
Important: For the **Network** setting, leave the default, which means the **Use network from current parent VM image** check box is selected.
11. With the vCenter Settings page completed, click **Next**.

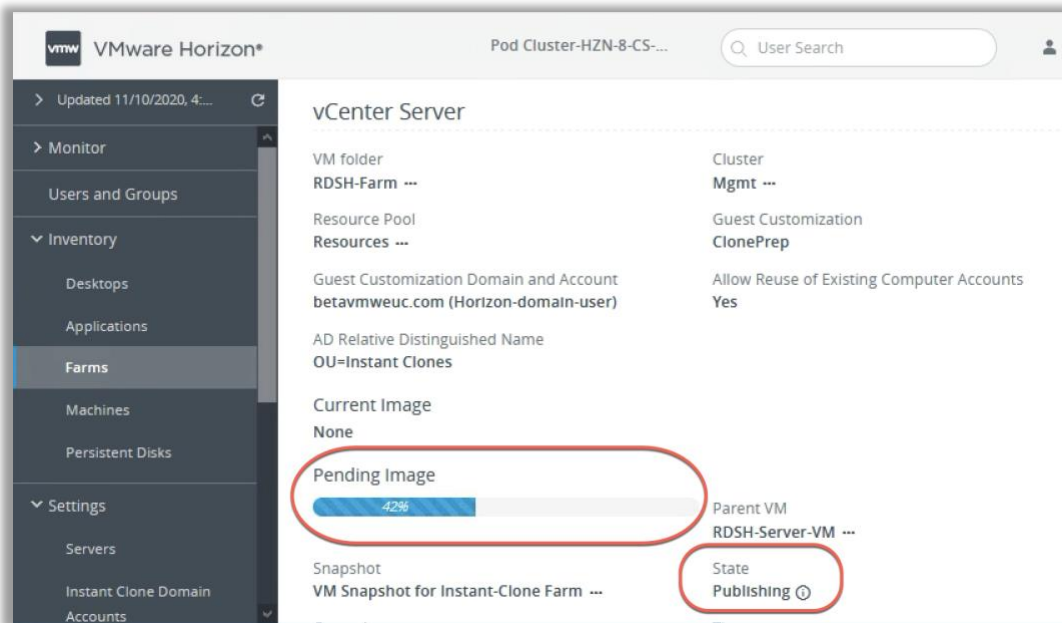
The screenshot shows the 'Add Farm - RDSH-Farm' wizard with the 'vCenter Settings' step selected in the left-hand navigation pane. The main content area is divided into three sections: 'Default Image', 'Virtual Machine Location', and 'Resource Settings'. Each section contains a list of settings with 'Browse' buttons for selection.

Section	Setting	Value	Action
Default Image	* Parent VM in vCenter	/DC/vm/Instant-Clone Desktops/RDSH-Server-VM	Browse
	* Snapshot	/VM Snapshot for Instant-Clone Farm	Browse
Virtual Machine Location	* VM Folder Location	/DC/vm/Instant-Clone RDSH Servers	Browse
Resource Settings	* Cluster	/DC/host/Mgmt	Browse
	* Resource Pool	/DC/host/Mgmt/Resources	Browse
	* Datastores	1 selected	Browse

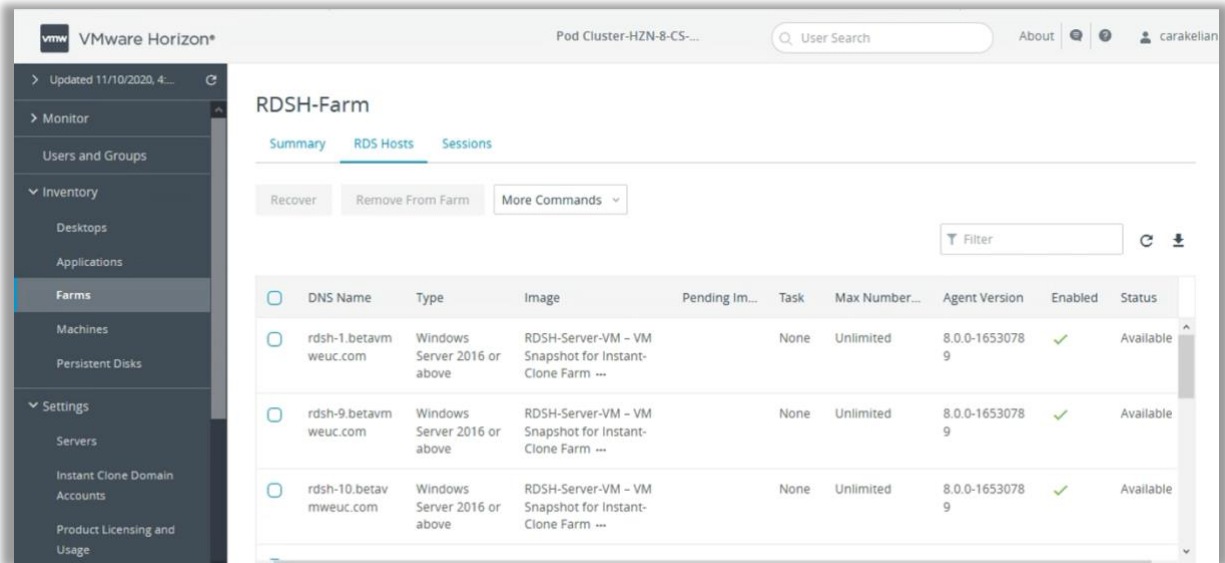
At the bottom right of the wizard, there are three buttons: 'Cancel', 'Previous', and 'Next'.

12. Complete the Guest Customization page, as follows:
 - a. Verify that the correct domain and domain admin account are selected.
This is the account you created and added as part of the prerequisites for this exercise.
 - b. For **AD container**, click **Browse** and select the OU that you created as part of the prerequisites for this exercise.
 - c. Leave the other default settings, and click **Next**.
13. On the Ready to Complete page, click **Submit**.
You are returned to the **Inventory > Farms** page.

14. To monitor farm creation, click the name of the farm in the list.
If you do not see the name in the list, click the Refresh icon above the table.
Because the VM image must be published before the instant clones can be created, the process might take several minutes.
15. On the farm details page for the instant-clone farm, scroll down to the **vCenter Server** section, and note the **State** field and the **Pending Image** field.
The **Pending Image** field shows the progress of the publish operation. When the farm is finished being created, the **State** field changes from **Publishing** to **Published**.



Note: The **Summary** tab is shown in this screenshot. To see details of the particular instant clone you just created, click the **RDS Hosts** tab.



Publish Applications Hosted on RDSH Servers

The published applications feature supports a wealth of remote-experience features, which include client-drive redirection, access to locally connected USB devices, file-type association, Windows media redirection, content redirection, printer redirection, location-based printing, 3D rendering, smartcard authentication, and more.

After applications are published, end users launch Horizon Client, or the HTML Access web client, to access a catalog of published applications. Selecting an application from the catalog opens a window for that application on the local client device, and the application looks and behaves as if it were locally installed.

For example, on a Windows client computer, an item for the application appears in the taskbar and looks identical to the way it would look if it were installed on the local Windows computer. Users can also create shortcuts for published applications, and the shortcuts appear on the client desktop, just like shortcuts for locally installed applications.

To publish applications, administrators create an application pool. Horizon automatically enumerates the installed applications on the RDSH servers. Administrators can select which of the applications to deploy and entitle users to.

Prerequisites for Publishing Applications

To perform this exercise, you need to have completed the exercise [Create an Instant-Clone RDSH Server Farm](#). Although it is possible to actually create the RDSH server farm as part of using the Add Pool wizard, the steps in this exercise direct you to select an existing server farm.

Run the Add Application Wizard to Create an Application Pool

1. In the Horizon Console, navigate to **Inventory > Applications**, click the **Add** button, and select **Add from Installed Applications**.
For this exercise, you will use installed applications. For information about adding an application pool manually, see [Worksheet for Creating an Application Pool Manually](#) in the guide *Setting Up Published Desktops and Applications in Horizon*.
Note: The format of the URL for accessing the web-based console is:
`https://<connection-server-FQDN>/admin`.
2. For **Application Pool Type**, leave the default, which is RDS Farm.
The server farm you created in the previous exercise should be displayed in the drop-down list.
3. Complete the Select Applications page, as follows, before clicking **Next**.
 - In the **Select installed applications** list, select the check boxes for the desired applications. In this example, we selected **Calculator** and **Paint**. A separate application pool will be created for each application you select. The list of applications includes both natively installed apps and App Volumes AppStacks or packages, if any, that you have attached to the servers.
 - De-select the **Entitle Users After Adding Pool** check box. You will entitle users in a later exercise.

Add Application Pool

1 Select Applications

2 Edit Applications

Select Installed applications

Name	Path
☐ h Recognition.Ink	
☒ Calculator	C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Accessories\Calculator.Ink
☐ Math Input Panel	C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Accessories\Math Input Panel.Ink
☒ Paint	C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Accessories\Paint.Ink

☐ Pre-Launch

Connection Server Restrictions

None

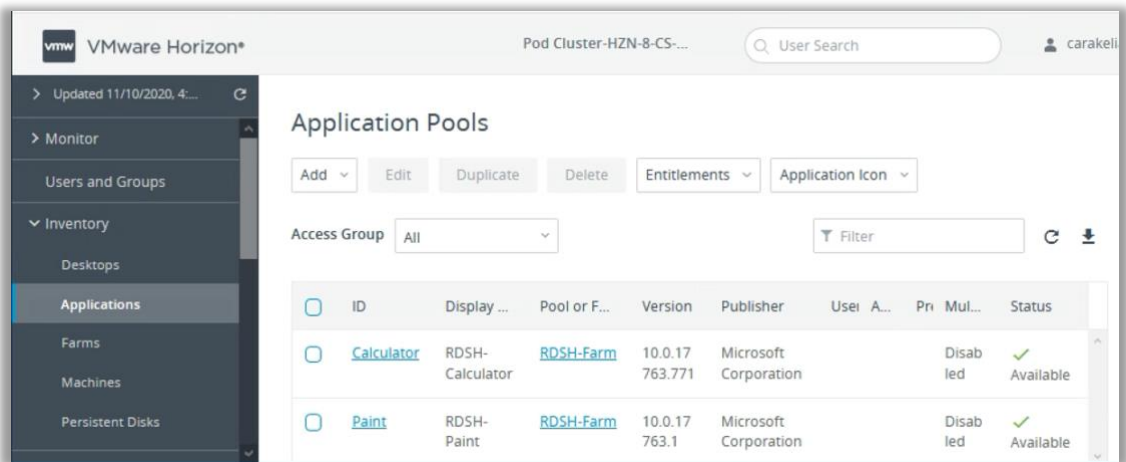
Category Folder

None

☐ Client Restrictions

☐ Entitle Users After Adding Pool

4. On the Edit Applications page, add **RDSH-** to the beginning of the display name. This way, if you later open the published app on a Windows computer, you will be able to distinguish between the locally installed app and the RDSH-published app.
5. Click **Submit**.
You are returned to the **Inventory > Applications** page, called Application Pools.



We used the default settings in most cases. For details about all the settings, see [Worksheet for Creating an Application Pool Manually](#) in the guide *Setting Up Published Desktops and Applications in Horizon*.

Performing Maintenance on a Server Farm

When you use automated instant-clone RDSH server farms, you can rapidly change the size of the farm, refresh the servers back to their original state and disk size, or update the servers to use a new golden image. Performing maintenance on an instant-clone farm means deleting the VMs in the farm and either recreating them from the current golden image or creating VMs from a new golden image, or snapshot.

- Administrators create a recurring maintenance schedule to restore the operating system disk of each VM in the farm to its original state and size, reducing storage costs. The VM is deleted and recreated from the currently selected golden image.
- Administrators schedule immediate maintenance to change the golden image used by the VMs in the farm, such as to apply an urgent security patch. This operation is similar to pushing a new VM image to a desktop pool, which you did in the exercise [Push a New Image to an Instant-Clone Desktop Pool](#).

You can use both types of schedules at the same time, and if you specified a minimum number of provisioned servers to be available during maintenance operations, your end users might never have their work interrupted.

Performing maintenance on a server farm is beyond the scope of this quick-start guide. For instructions, see [Managing Farms](#) in the guide *Setting Up Published Desktops and Applications in Horizon*.

Provisioning Users and Accessing Desktops and Apps

The first part of this chapter walks you through the process of entitling end users to a desktop or application pool. The second part of this chapter shows you how to connect to a virtual desktop or published application as an end user would.

Entitling Users

You can entitle users to an application pool or desktop pool either at the time you create the pool or after the pool is created.

- To entitle users when you create the pool, at the end of the Add Application Pool wizard or Add Desktop Pool wizard, you can select the **Entitle users after this wizard finishes** check box.
- To create user entitlements after the pool is created, for application pools, you can select multiple application pools, and entitle users to all the selected pools. For desktop pools, you must select one pool at a time.

It is also possible to set up the system so that end users can access RDSH application pools without having to authenticate at all. For more information, see [Configuring Unauthenticated Access](#) in the *Horizon Administration* guide.

Note: For this exercise, you create local entitlements, which entitle users to desktops within one Horizon pod. A pod is a group of interconnected Connection Servers running in the same LAN segment that broker desktops or published applications. For information about using the Cloud Pod Architecture feature to create global entitlements, which entitle users to multiple desktops across multiple pods in a pod federation, see the guide [Administering Cloud Pod Architecture in Horizon](#).

Prerequisites for Entitling Users

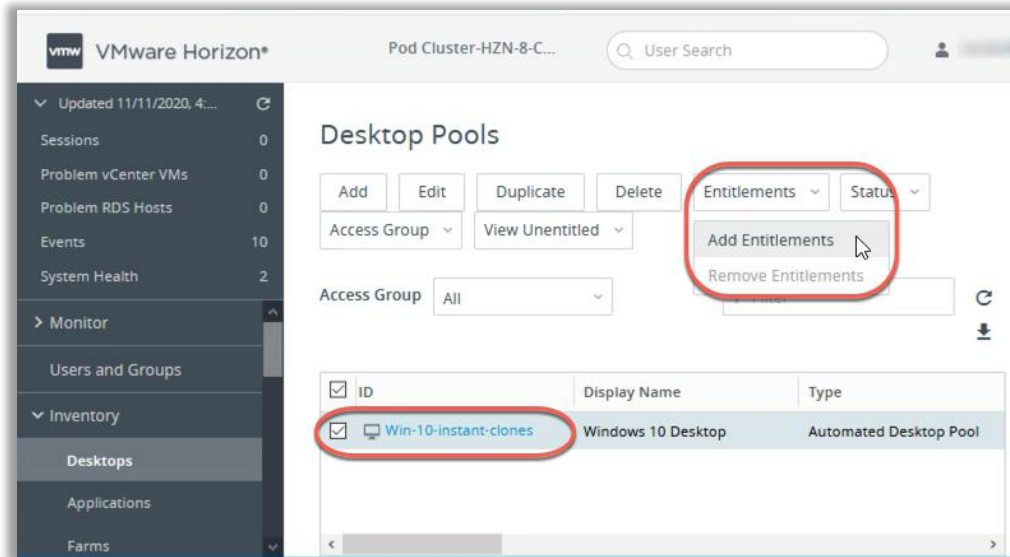
Before you can entitle users, you must have created a desktop or application pool. Exercises for performing these tasks are included in the chapters [Creating Single-User Desktop Pools](#) and [Creating RDSH-Published Desktops and Applications](#).

Entitle End Users to Application Pools or Desktop Pools

Entitling users means specifying which users and groups are allowed to access the desktop or application. For this exercise, you create user entitlements after the pool is created.

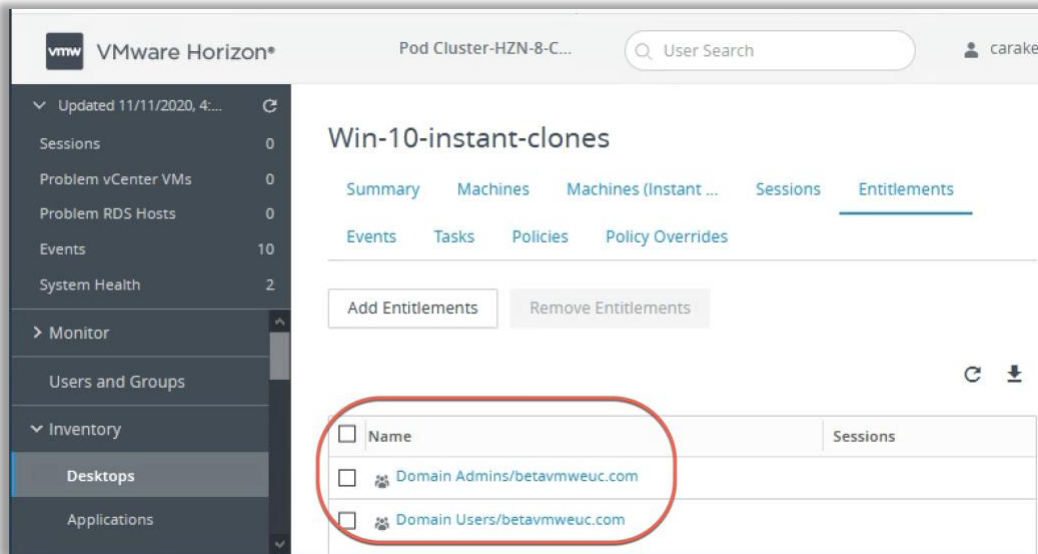
1. Start the Add Entitlements wizard for the desired desktop pool or application pools, as follows:
 - a. In the Horizon Console, and navigate to **Inventory > Desktops** or, for application pools, navigate to **Inventory > Applications**.
The format of the URL for accessing the console is:
`https://<connection-server-FQDN>/admin`
 - b. Select the check box next to the name of the pool you want to entitle users to.
Important: If you are entitling users to application pools, you can select multiple pools, and entitle users to all the selected pools. For desktop pools, you must select one pool at a time.

- c. Click the **Entitlements** button and select **Add Entitlements**.



2. In the Add Entitlements dialog box, click **Add**.
3. Use the Find User or Group dialog box to search for users.
For example, for this exercise, you could select **Starts with** and enter a Dso that all user and group names that begin with *D* will be returned, such as Domain Users. If you leave the text boxes empty, all users and groups are returned.
4. From the list of users and groups returned, select the users or groups to entitle, and click **OK**.
5. In the Add Entitlements dialog box, verify that the desired users or groups now appear in the list, and click **OK**.
Note: The **Add** button in this dialog box is for adding additional users to the list. The check boxes are for selecting a user or users you want to remove.
You are returned to the Application Pools page or the Desktop Pools page.

- To verify that the entitlements have been added, click the name of the desktop or application pool in the list of pools, and select the **Entitlements** tab.



Note: You can also use the buttons on the **Entitlements** tab to add and remove user entitlements for a specific pool.

Launching Remote Desktops and Applications from Client Devices

After you have finished deploying virtual desktops or published applications and entitling users, you are ready to explore end-user connection options. End users can connect to desktops and applications using different Horizon Clients, including desktop and mobile clients. VMware provides native Horizon Clients for iOS, Android, Chrome, macOS, Windows, and Linux.

Alternatively, you can use the HTML Access web client by entering the URL of your Connection Server, using the following format:

`https://<FQDN or IP address>`

On the VMware Horizon web portal page that appears, you can click either the icon that takes you to the Horizon Clients download page or the icon for logging in using the HTML Access web client.

Prerequisites for Connecting to a Desktop or Application with Horizon Client

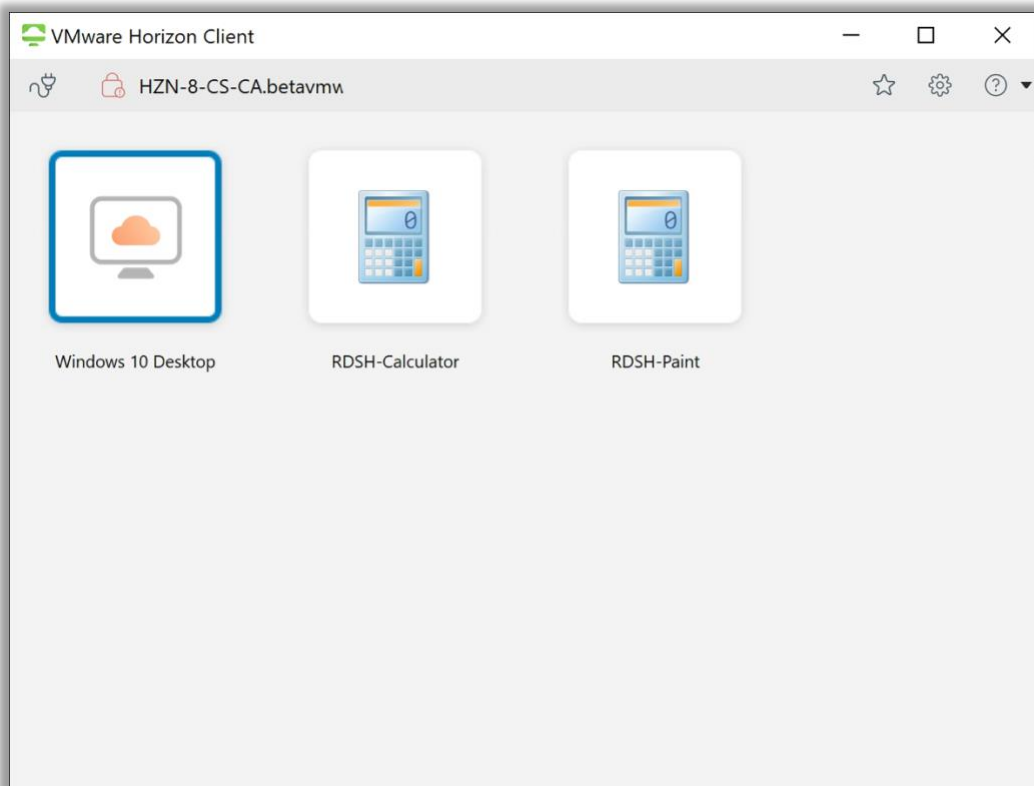
To perform this exercise, you need the following:

- **Endpoint PC** – You can use a Mac, Linux, or Windows PC.
- **Horizon Client installed** – Go to the [Download VMware Horizon Clients](#) page, and download and install the free Horizon Client software.
- **User account with admin privileges** – To install the Horizon Client software, you must log in to the endpoint device as a user with administrative privileges.
- **Connection Server address** – Verify that you have the fully qualified domain name of the Connection Server that brokers connections to the desktop and application pools you created in earlier exercises.

- **Desktop or application pools** – Exercises for creating pools are included in the chapters [Creating Single-User Desktop Pools](#) and [Creating RDSH-Published Desktops and Applications](#).

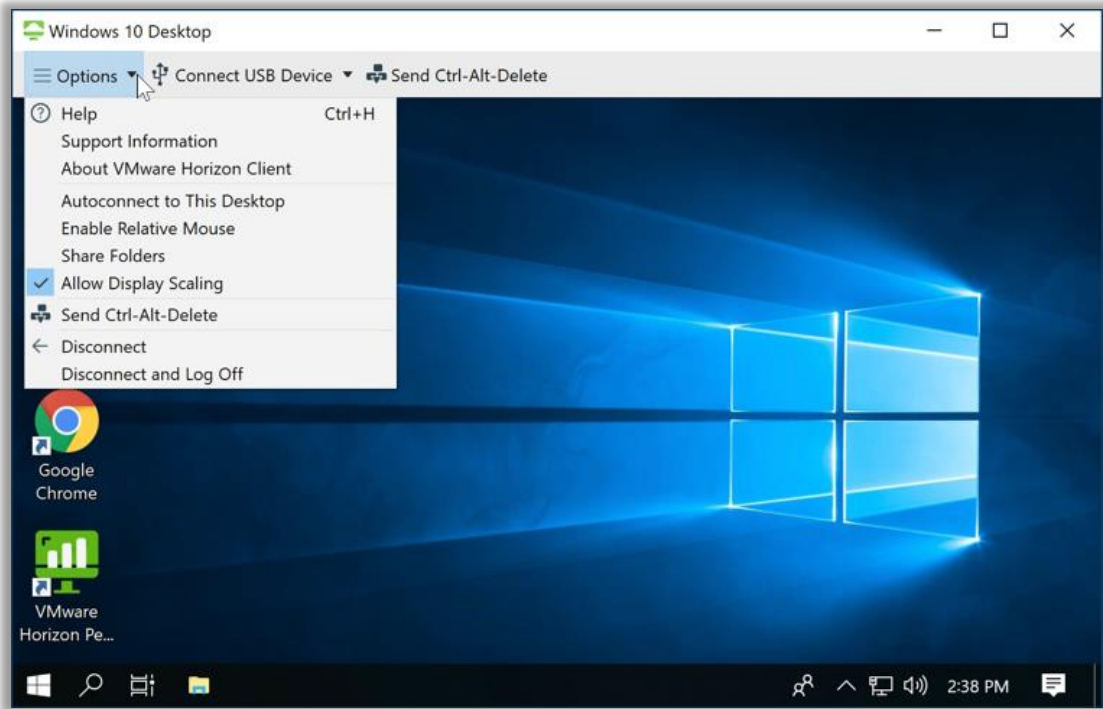
Launch a Virtual Desktop or Published App

1. On your client computer, start VMware Horizon Client the same way you would start any application. For example, on a Windows PC, double-click the desktop icon.
2. In the VMware Horizon Client window, click the **New Server** button.
3. When prompted, enter the FQDN of the Connection Server, and click **Connect**.
4. If you receive a security warning, click **Continue** to bypass the certificate warning. If you install a CA-signed security certificate on the machine that hosts the Connection Server, this warning does not appear.
5. In the Login dialog box, enter the user name and password of a user who is entitled to the desktop or published applications, and click **Login**.
6. To launch an application or desktop, double-click the icon for the application or desktop.



7. In the Sharing dialog box, click **Allow** to allow access to files on your client device, as well as locally connected storage devices such as USB thumb drives, while using virtual desktops and published applications.

8. Verify that you have successfully logged in to your desktop or application.
For this example, we have successfully logged in to an instant-clone VM from the Windows 10 Desktop pool.



This screenshot also shows the list of options available from the Horizon Client menu bar.

9. To disconnect from the virtual desktop or published app, click the **X** in the upper-right corner, as you would for any other Windows application or window, and, for desktops, confirm that you want to disconnect.

This exercise showed you how to use Horizon Client on a desktop or laptop computer. You can also use Horizon Client on a tablet or phone. For OS-specific instructions, go to the [VMware Horizon Client Documentation](#) page.

Use the HTML Access Web Client

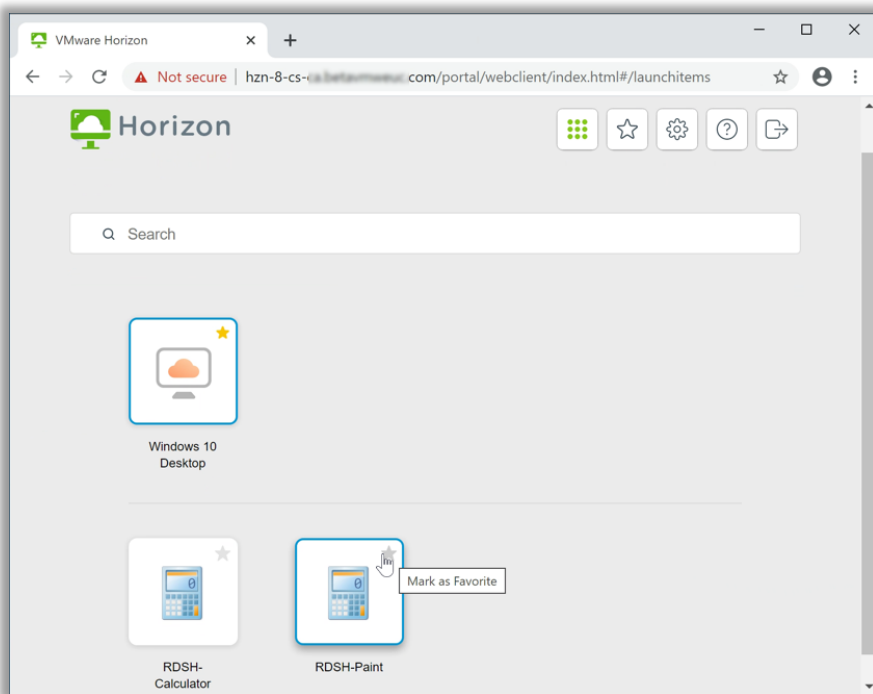
You can connect to virtual desktops and published applications from an HTML5-enabled web browser. All the latest browsers are supported, and are listed in the [VMware Horizon HTML Access User Guide](#).

To perform this exercise, you need the following:

- **Connection Server address** – Verify that you have the fully qualified domain name of the Connection Server that brokers connections to the desktop and application pools you created in earlier exercises.
- **Desktop or application pools** – Exercises for creating pools are included in the chapters [Creating Single-User Desktop Pools](#) and [Creating RDSH-Published Desktops and Applications](#).
- **Chrome browser** – (Optional) To display the **Allow H.264 decoding** setting, which is pictured in one of the following steps, you must use a Google Chrome browser.

To access a virtual desktop or published app by using HTML Access:

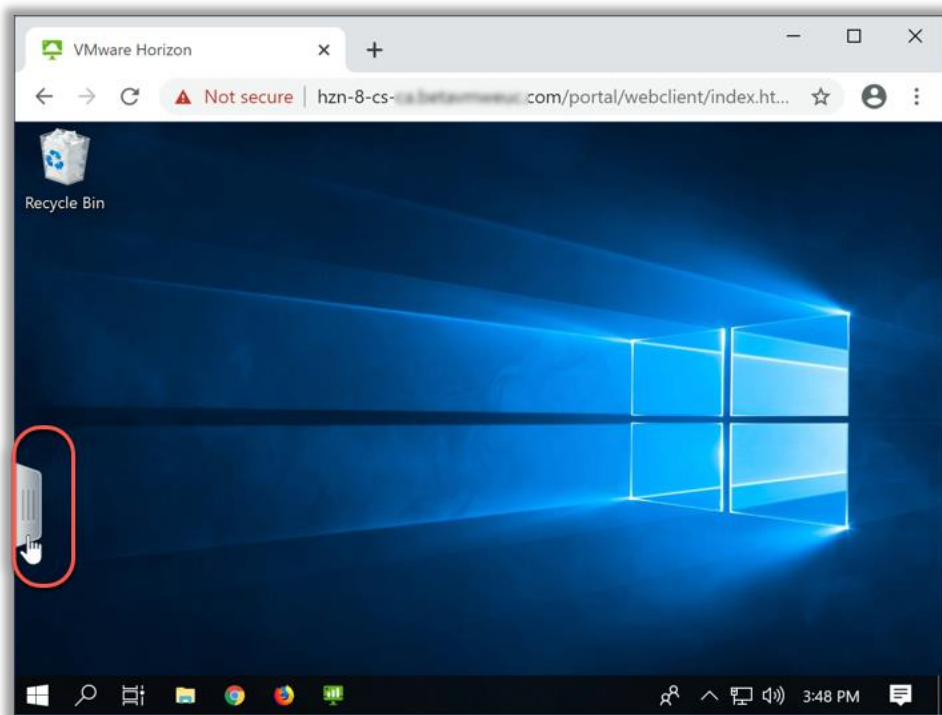
1. Open a supported web browser and enter the address of your Connection Server. The URL format is `https://<connection-server-FQDN>`
Note: If you do not have a CA-signed security certificate, you might be prompted to add a security exception to your browser.
2. When the VMware Horizon page appears, click **VMware Horizon HTML Access**.
3. Enter credentials of a user who is entitled to the desktop or application pool, and click **Login**.
 After the credentials are validated, you can see the available desktops and applications.
4. To mark a desktop or application shortcut as a favorite, click a star in the corner of the icon.



This feature is convenient if you have many desktops and applications and do not want to have to scroll to find the applications and desktops you use most frequently. You can also use the **Search** box to quickly locate an application or desktop if you know its display name.

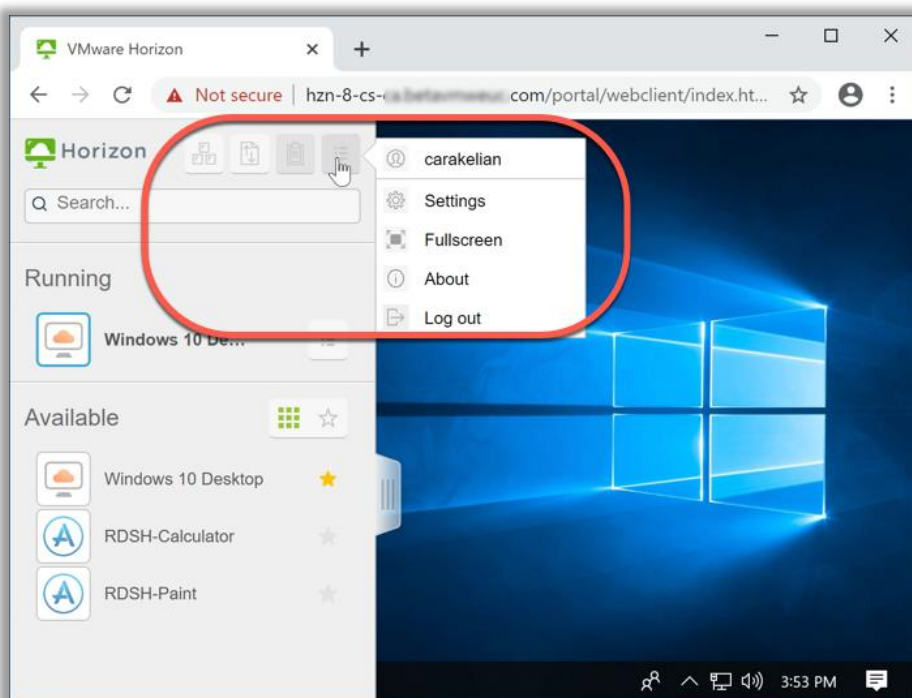
5. Click the desktop icon to display the virtual desktop in your browser.

6. Click the tab on the left side of the screen to open the navigation sidebar.

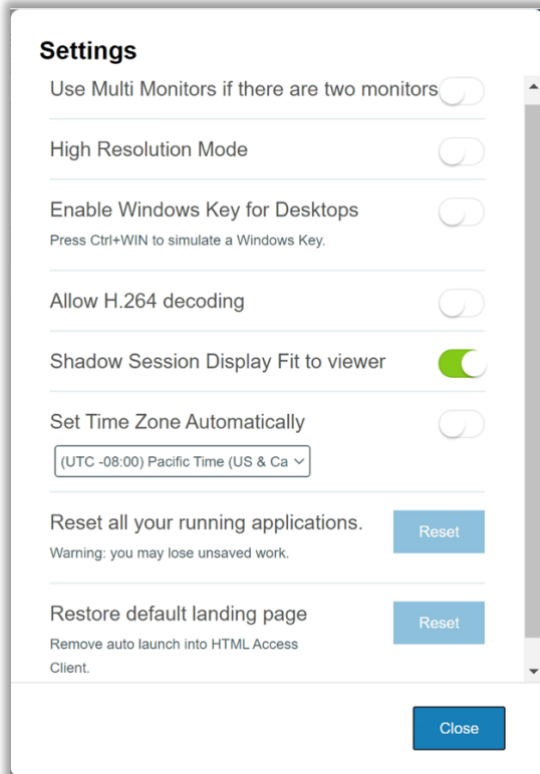


7. Hover your cursor over each toolbar button to display its tooltip. You can use the toolbar at the top of the sidebar to:
- Send Ctrl+Alt+Del to the application work area
 - Transfer files, if the feature is enabled
 - Open the Copy & Paste panel
 - Open the Settings menu

8. Click the **Open Menu** icon on the right end of the toolbar, and select **Settings**.



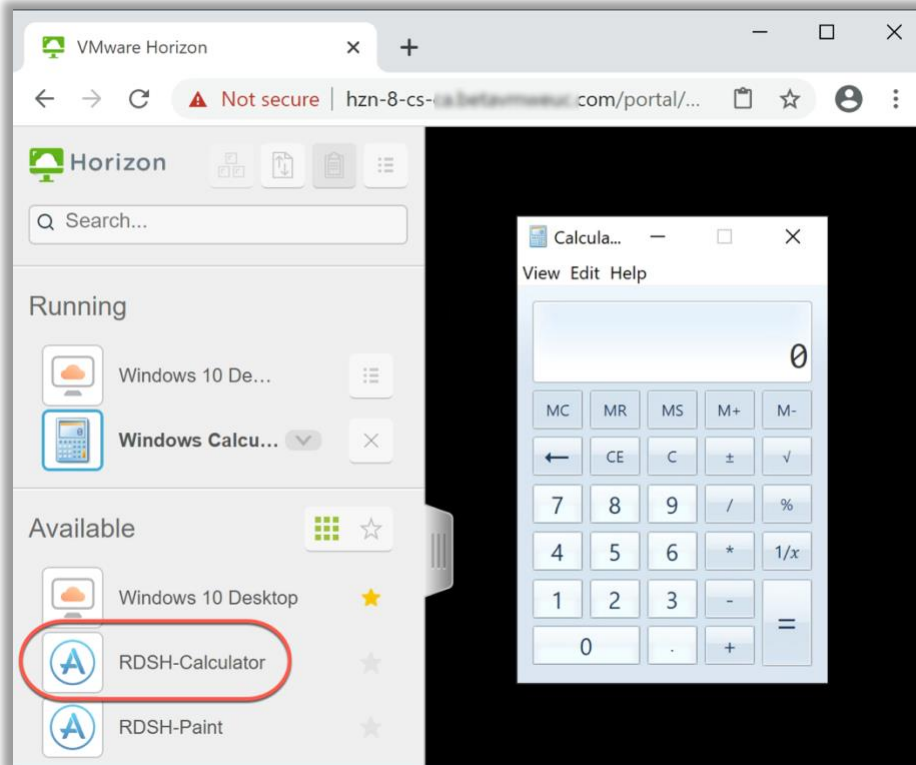
9. Review all the settings available to provide a rich user experience, and then click **Close**.



When you use a Chrome browser and use the VMware Blast Extreme display protocol, the **Allow H.264 decoding** setting causes the graphics processor on the client device to do the work involved in playing back video and images. Hardware decoding offloads the work to the GPU, so that CPU consumption is reduced, resulting in less device power consumed, for longer battery life. To make the setting take effect, you must disconnect and reconnect to the desktop or application.

For information about the **Shadow Session Display Fit to viewer** setting, see [Sharing Remote Desktop Sessions](#).

10. Click an application in the sidebar to launch it.
For this example, we opened the Calculator app.



Note: In the sidebar, you can click the star icon to the right of an application or desktop name to designate the item as a favorite, and click the star above the list to display only favorites.

11. From the **Running** list, which displays the names of running desktops and applications, click the **Menu** toolbar button next to the desktop name. You can then select **Close**, which disconnects you from the desktop, or select **Log off**.

This exercise described using the HTML Access web client, which does not require installing any software on the client device. For information about HTML Access features such as copying and pasting or transferring files between your local client system and the virtual desktop or published application, see the [HTML Access documentation](#).

Summary and Additional Resources

This quick-start tutorial demonstrated just how quickly and easily you can use VMware Horizon to create VDI desktops and RDSH-published applications using a Horizon-on-vSphere infrastructure. You completed simple wizards to install and configure a Connection Server, which streamlines provisioning of RDSH servers and cloned desktops.

You then created an automated desktop pool and an automated RDSH server farm. With one simple wizard, you created multiple application pools. Next, you entitled end users to applications and desktops. In addition, this guide provided an overview of features, architecture, and components.

Finally, you enjoyed the end-user experience of launching desktops and published applications from the Windows-based Horizon Client and the web-based HTML Access client. The native Horizon Client software can be installed on Windows, macOS, Linux, Chrome, iOS, and Android endpoint devices.

Because this guide is meant to get you started quickly, it does not delve into details of all the options and features that provide a rich user experience:

- Support for use cases such as graphics-intensive 3D applications with NVIDIA GRID vGPU and Unified Communications with Microsoft Skype for Business
- Quick and easy access to a user's files from their virtual desktops and applications with file-type association
- Support for the most commonly used peripherals, including printers, scanners and imaging devices, smart cards, and USB storage devices
- Performance optimizations to increase application responsiveness

For information about these and other important features, see the items listed in the [Additional Resources](#) section.

Additional Resources

[VMware Horizon Documentation](#)

[Horizon Resources on Digital Workspace Tech Zone](#)

[Mastering Horizon – Tech Zone Activity Path](#)

[Quick-Start Tutorial for VMware Horizon Cloud Service on Microsoft Azure](#)

Changelog

The following updates were made to this guide.

Date	Description of Changes
2021-01-05	Added a contributor.
2020-12-08	Initial publication date.

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